

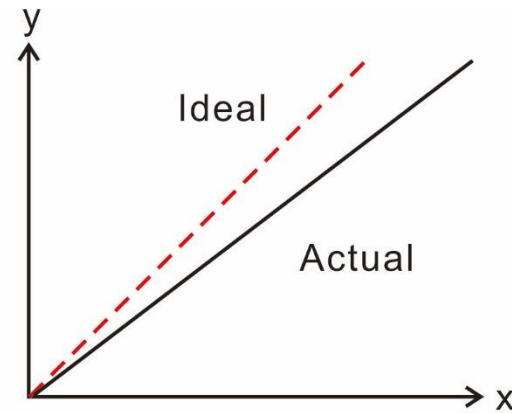
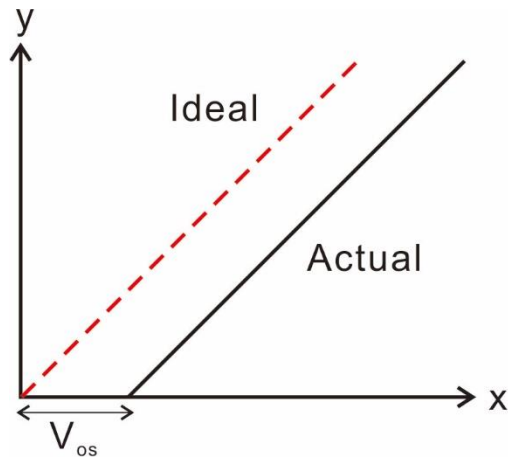
Data Converter Performance Metrics

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Static Performance Metrics

- Offset
- Gain error
- Differential nonlinearity (DNL)
- Integral nonlinearity (INL)

Basic Concept

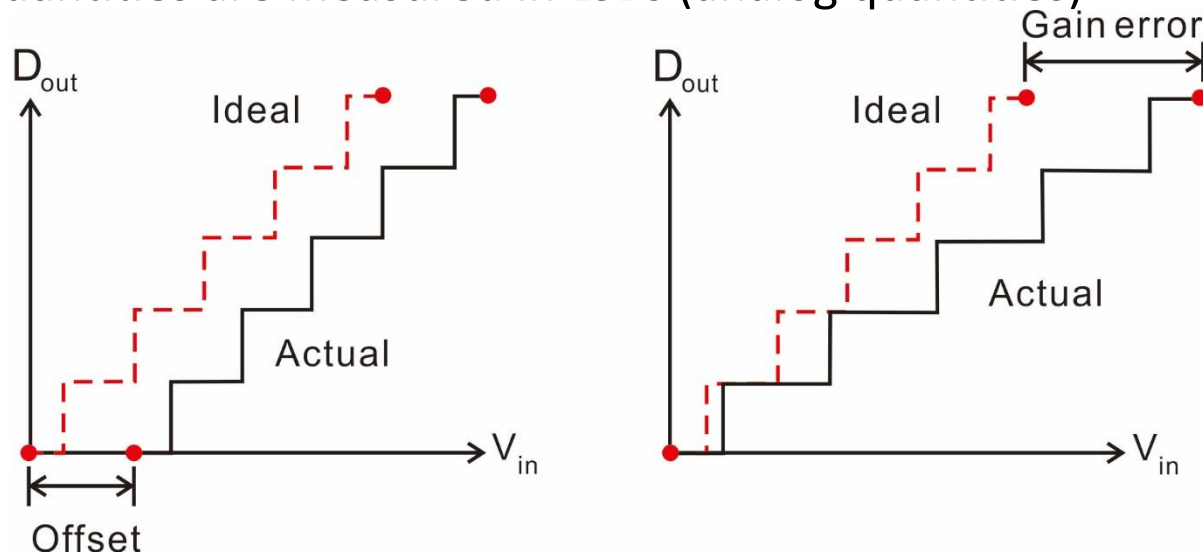


$$\begin{aligned} y &= ax + b \\ &= x + (a - 1)x + b \end{aligned}$$

$\underbrace{\hspace{1.5cm}}$ $\underbrace{\hspace{1.5cm}}$
Gain error Offset error

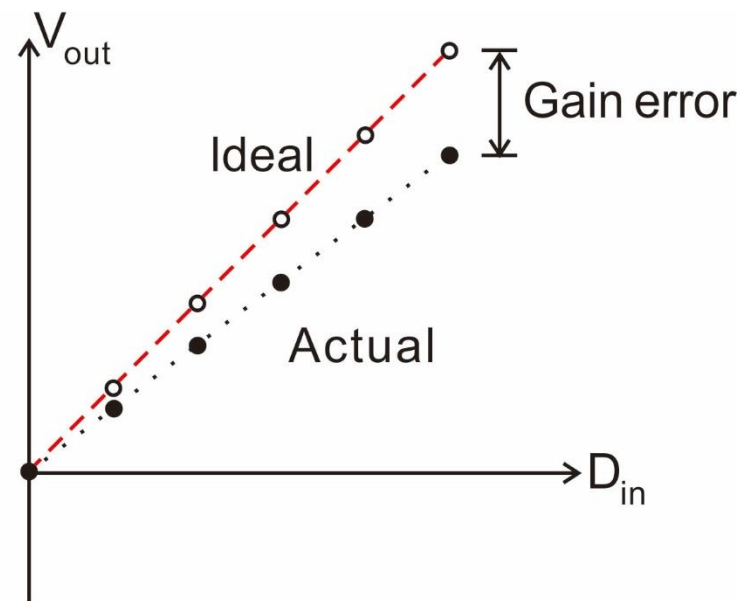
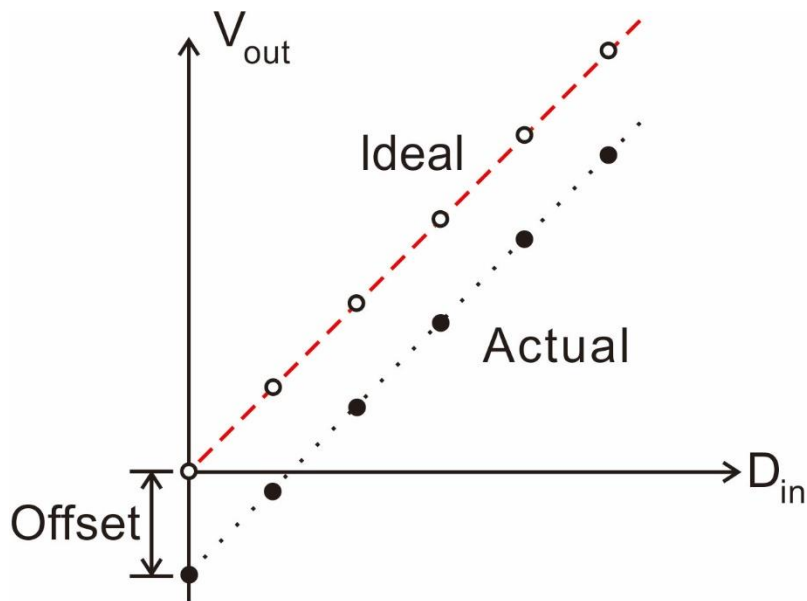
ADC Offset and Gain Error

- Start point: $\frac{1}{2}$ LSB before first transition
- End point: $\frac{1}{2}$ LSB after last transition
- **Offset**: the deviation of **start point** from its ideal location
- **Gain error**: the deviation of **end point with offset removed**
- Both quantities are measured in LSBs (analog quantities)



DAC Offset and Gain Error

- Start and end points are defined by **analog output values** corresponding to the smallest and largest digital inputs.
- Offset and gain errors are also measured in LSBs (analog quantities)



Comments

- Definitions in the previous slides are widely used in industry
 - Note that different companies may use different definitions
 - Read datasheet carefully
 - IEEE standard defines them based on least square curve fitting
- Gain and offset errors matters when we care about absolute magnitude
 - E.g., in scientific measurements
- However, we may not care about offsets and gain errors in many applications
 - These errors are easy to correct in the digital domain
 - Offsets and gain errors do not affect linearity
 - E.g., audio application, communication, etc
- We are more interested in nonlinearities.

Differential Nonlinearity (DNL)

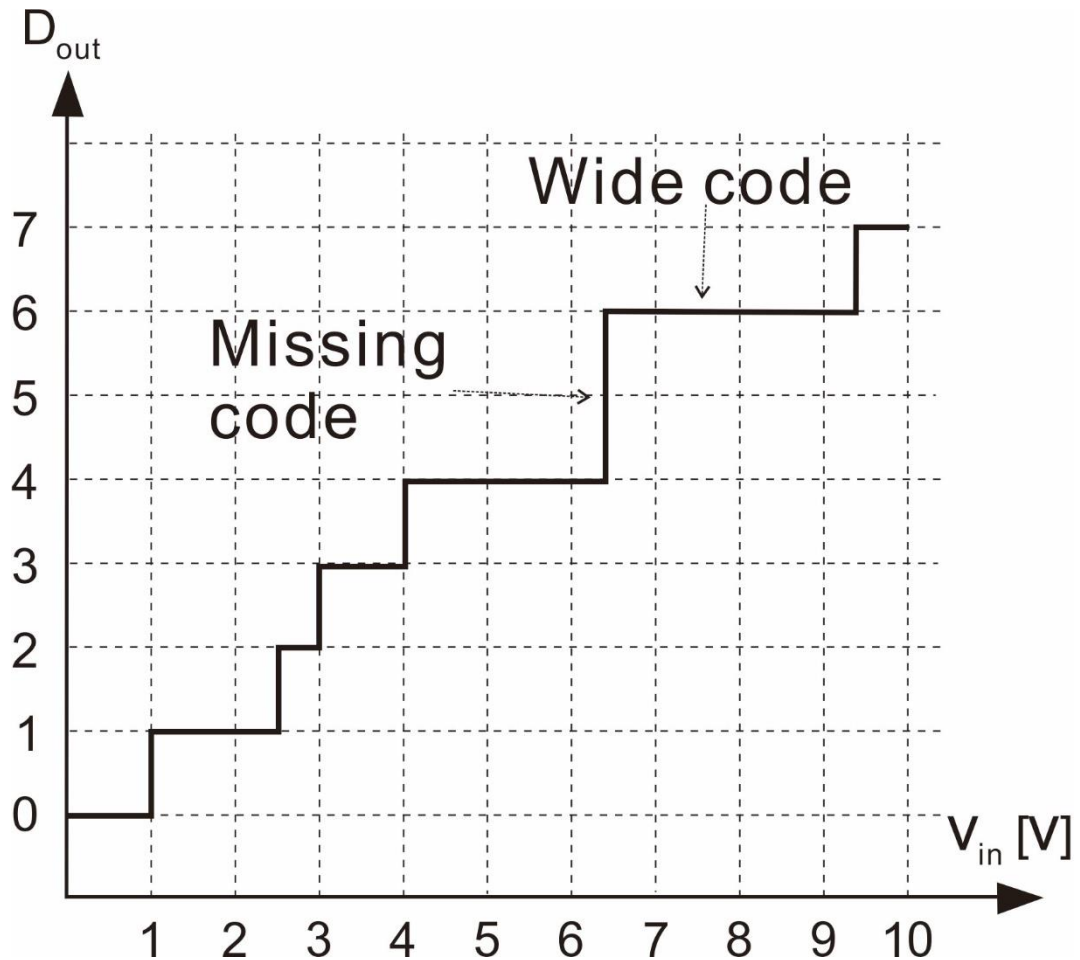
- For an ideal ADC, all codes have the same width.
- For an ideal DAC, all steps have the same size.
- However, in practice, the code widths and step sizes may not be identical.
- DNL[k] is defined as the deviation of the width from the "average" width for code k.
 - It is a measure of uniformity.
 - Gain and offset errors do not affect uniformity.

$$\text{DNL}[k] = \frac{W[k] - W_{\text{avg}}}{W_{\text{avg}}}$$

$$W[k] = T[k + 1] - T[k] \qquad W_{\text{avg}} = \frac{\sum_{k=1}^N W[k]}{N} = \frac{T[N + 1] - T[1]}{N}$$

ADC DNL Example

Code [k]	T[k]	W [V]
0	undefined	undefined
1	1.0	1.5
2	2.5	0.5
3	3	1
4	4	2.4
5	6.4	0
6	6.4	3
7	9.4	undefined



ADC DNL Example

- What is the average code width?
 - There are in total 6 codes (not counting the first and last codes)
 - The total code width is 8.4V

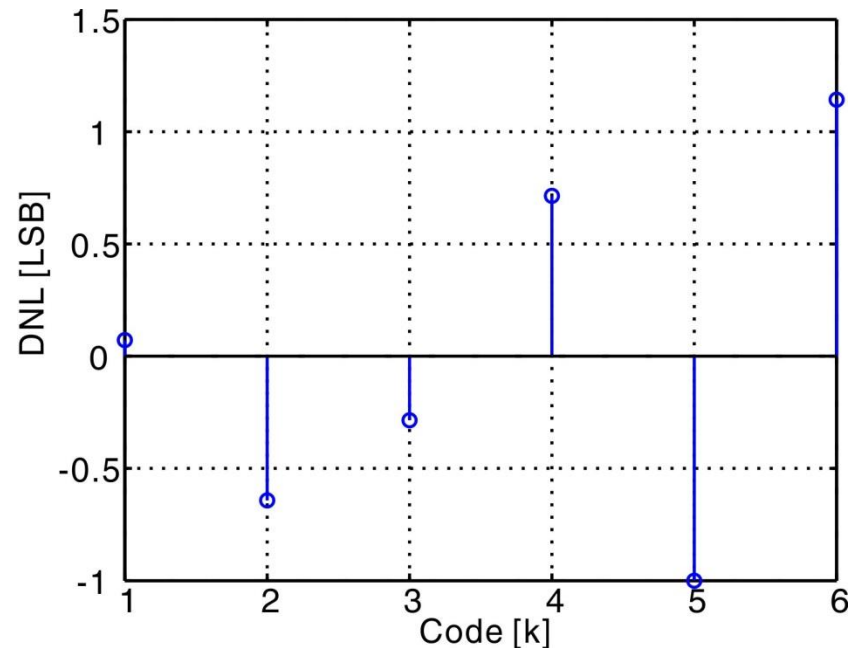
$$W_{\text{avg}} = \frac{9.4\text{V} - 1.0\text{V}}{6} = 1.4\text{V}$$

- DNL[k] is defined as

$$\text{DNL}[k] = \frac{W[k] - W_{\text{avg}}}{W_{\text{avg}}} = \frac{W[k] - 1.4}{1.4}$$

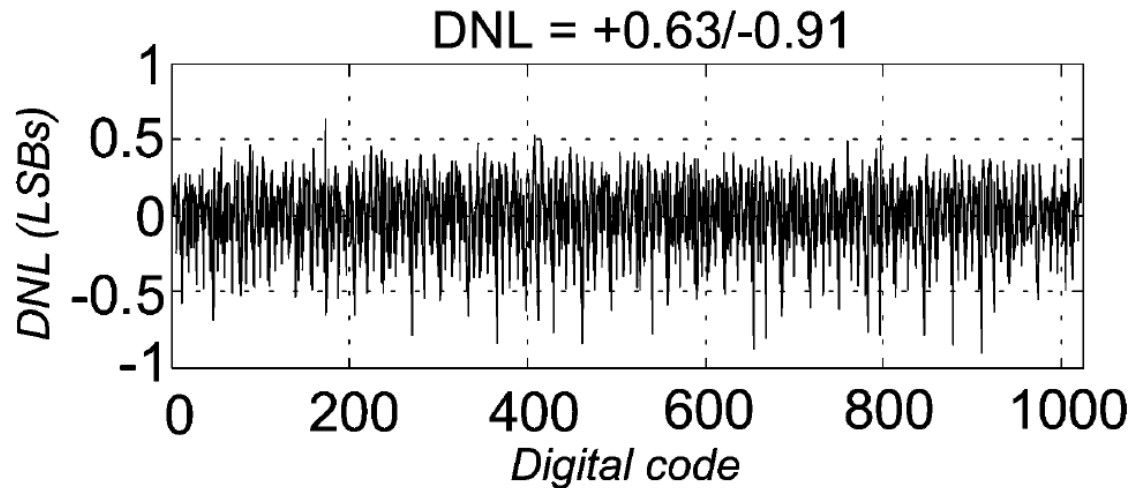
Result

Code [k]	DNL [LSB]
1	+0.07
2	-0.64
3	-0.29
4	+0.71
5	-1.00
6	+1.14



- Positive/negative DNL implies wide/narrow code
- DNL = -1 LSB implies missing code
- Possible to have DNL > +1 LSB
- Impossible to have DNL < -1 LSB for an ADC (why?)
- The sum of all DNL[k] is equal to zero (why?)

Typical ADC DNL Plot



[Ahmed, JSSC 12/2005]

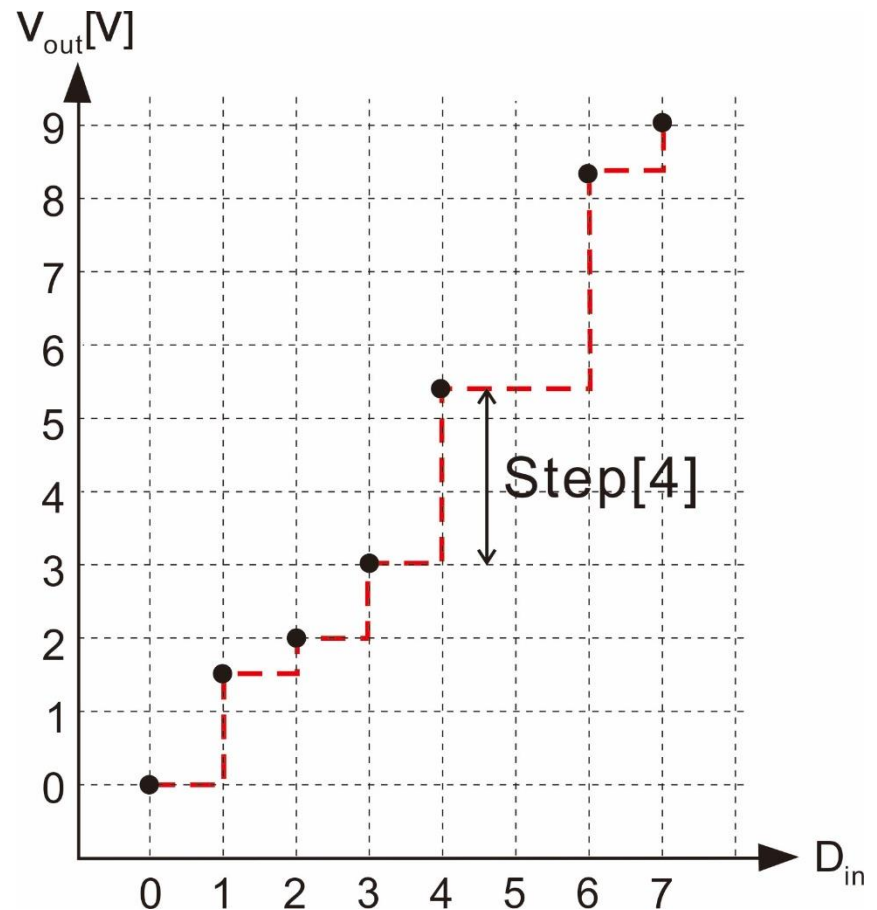
- Usually, we only specify the min/max DNL numbers across all codes
 - E.g. DNL = +0.63/-0.91 LSB
- A well designed ADC usually has DNL bounded by ± 1 LSB
- DNL of an ADC is usually obtained by using a linear ramp input and measuring the duration of all codes.

DAC DNL

- Same idea as that for an ADC.
 - Note that static metrics are all defined as analog values.

$$\text{DNL}[k] = \frac{\text{Step}[k] - \text{Step}_{\text{avg}}}{\text{Step}_{\text{avg}}}$$

- DAC DNL can be less than -1 LSB. Why?

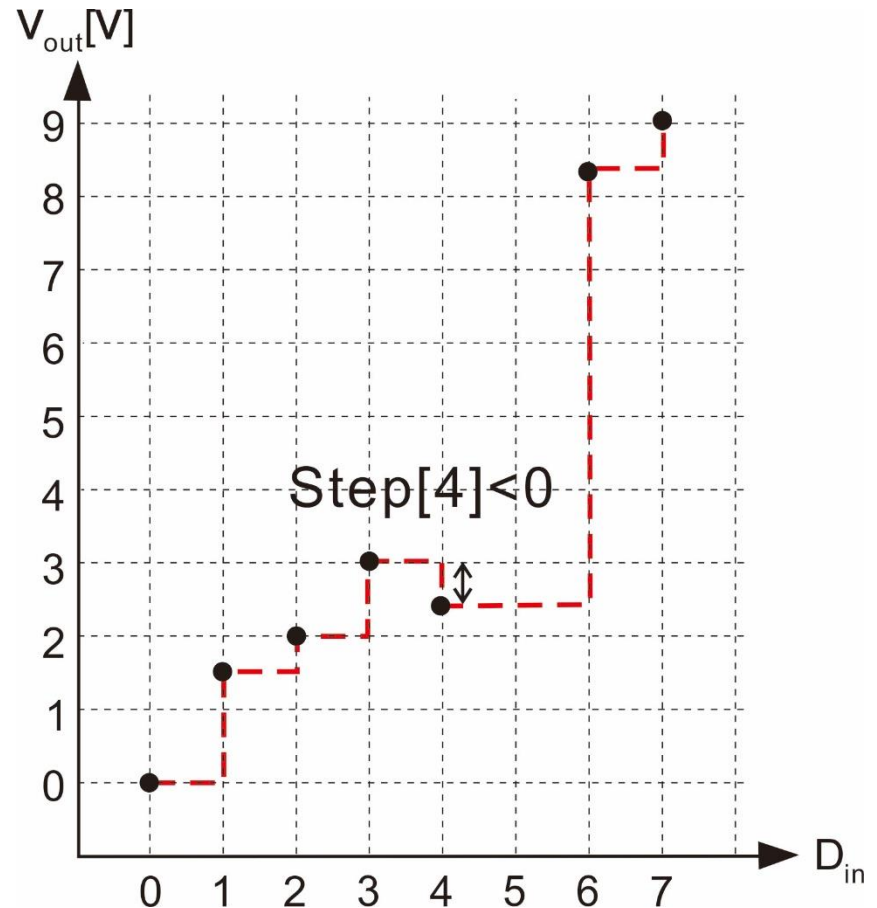


Non-Monotonic DAC

- $DNL < -1$ LSB implies non-monotonicity

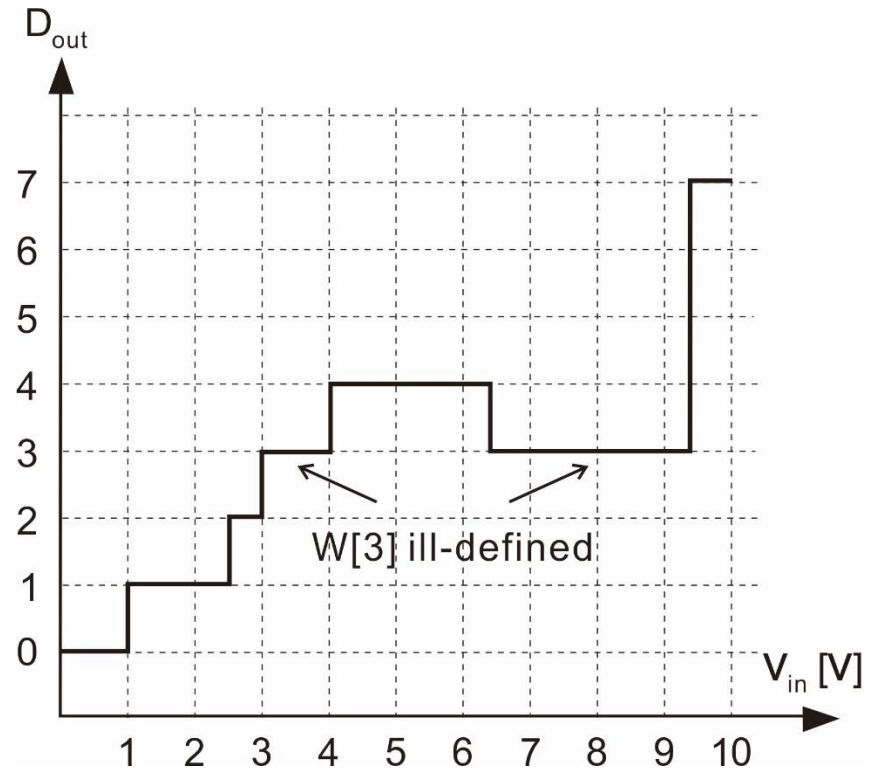
$$DNL[4] = \frac{Step[4] - Step_{avg}}{Step_{avg}} < -1 \text{ LSB}$$

- How about a non-monotonic ADC?



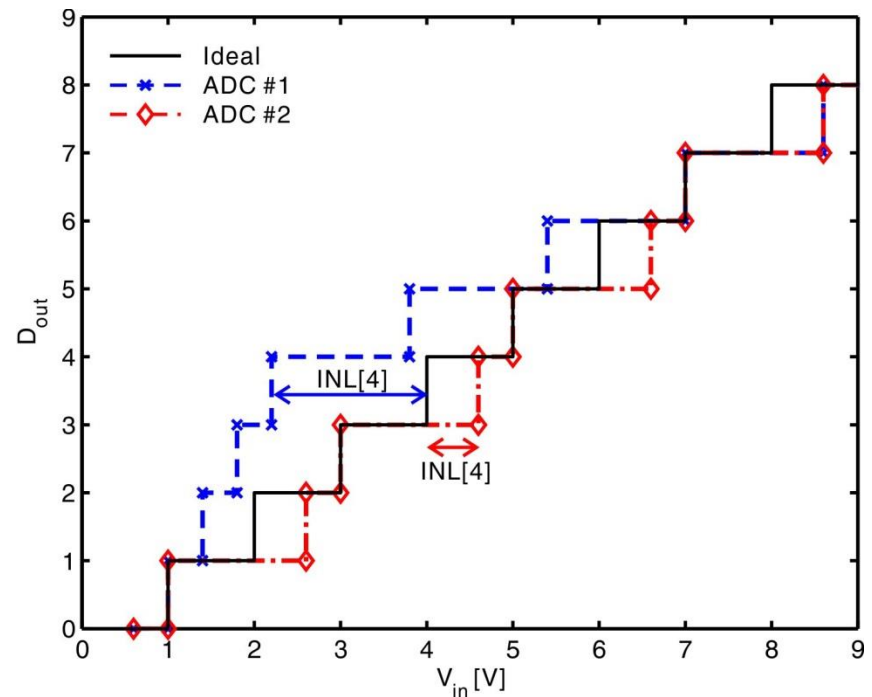
Non-Monotonic ADC

- Code 3 has 2 transition levels
- $W[3]$ is ill-defined, and thus, $DNL[3]$ is ill-defined.
- Not a major issue, because such a non-monotonic ADC will not be sold anyway.

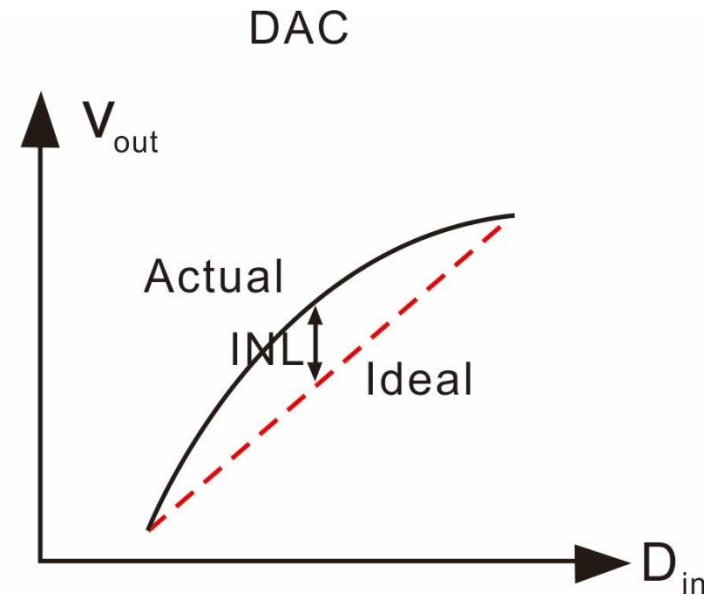
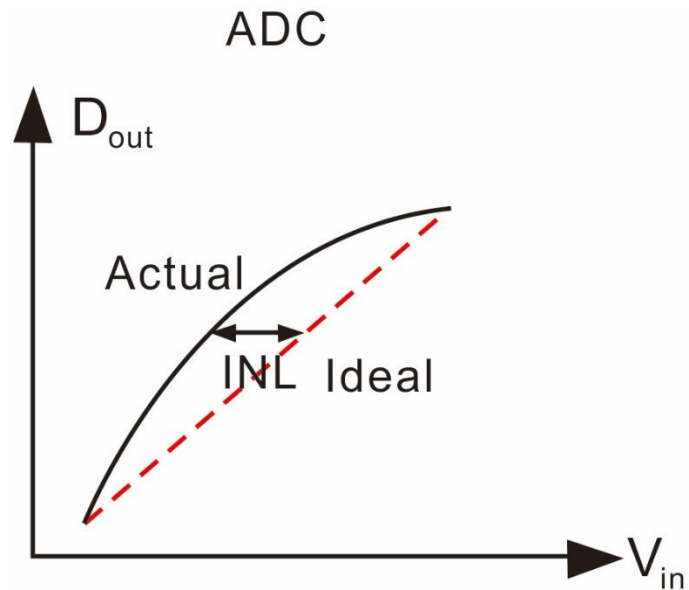


Integral Nonlinearity (INL)

- DNL, by itself, is insufficient for characterizing the nonlinearities.
- Compare the following two ADC, which one is more nonlinear?
 - ADC #1 DNL = $[-0.6, -0.6, -0.6, -0.6, +0.6, +0.6, +0.6, +0.6]$
 - ADC #2 DNL = $[-0.6, +0.6, -0.6, +0.6, -0.6, +0.6, -0.6, +0.6]$
- INL quantifies the overall deviation of the transfer curve from an ideal straight line.



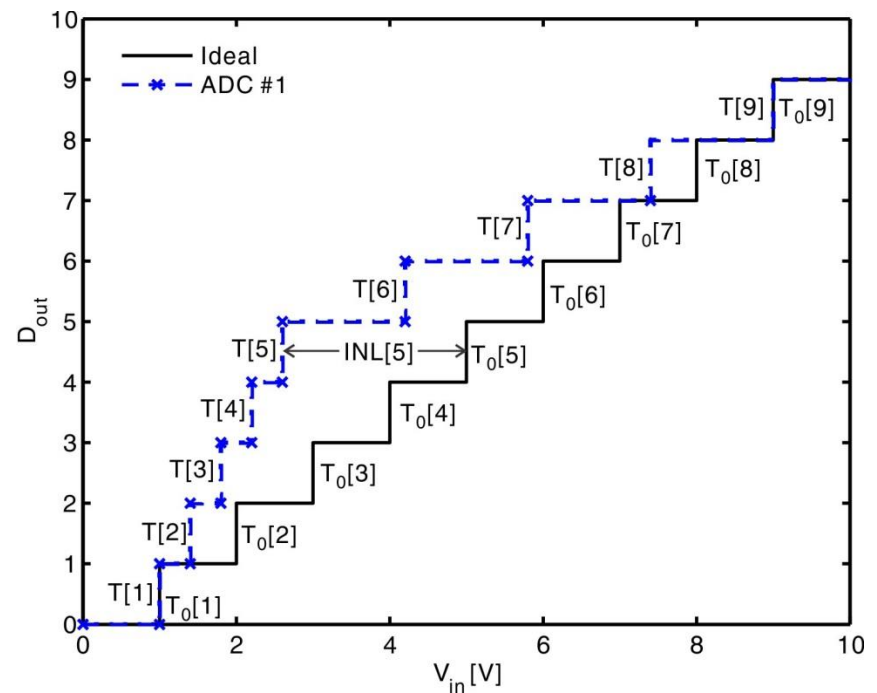
General Idea for INL



ADC INL Example

$$\text{INL}[k] = \frac{T[k] - T_0[k]}{W_{\text{avg}}}$$

- $T_0[k]$ is the ideal evenly spaced transition level
- $\text{INL}(0)$ is undefined
- $\text{INL}(1) = \text{INL}(9) = 0$
- Similar as DNL, INL is also independent from gain and offset errors



Relationship between INL and DNL

- It is easy to derive that:

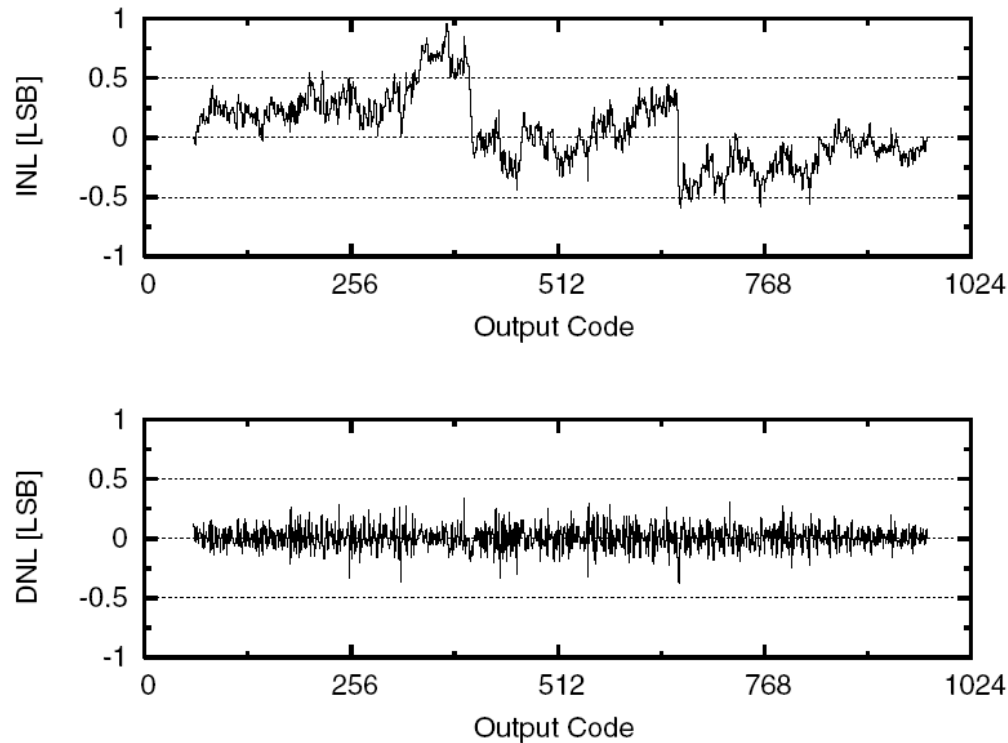
$$\text{INL}[k] - \text{INL}[k - 1] = \text{DNL}[k - 1] \quad \Rightarrow \quad \text{INL}(k) = \sum_{i=1}^{k-1} \text{DNL}(i)$$

- Once we know DNL, we can easily find INL.

	ADC #1		ADC #2	
Code [k]	DNL [LSB]	INL [LSB]	DNL [LSB]	INL [LSB]
1	-0.6	0	-0.6	0
2	-0.6	-0.6	+0.6	-0.6
3	-0.6	-1.2	-0.6	0
4	-0.6	-1.8	+0.6	-0.6
5	+0.6	-2.4	-0.6	0
6	+0.6	-1.8	+0.6	-0.6
7	+0.6	-1.2	-0.6	0
8	+0.6	-0.6	+0.6	-0.6
9	Undefined	0	Undefined	0

- ADC #1 is more nonlinear.

Typical ADC DNL/INL Plot



[Ishii, CICC, 2005]

- DNL/INL plots reveals architectural details.
- A well-designed ADC has both INL and DNL bounded by ± 1 LSB

DAC INL

- Same idea applies
 - Find ideal output values that lie on a straight line between endpoints
 - Calculate INL for each code k using

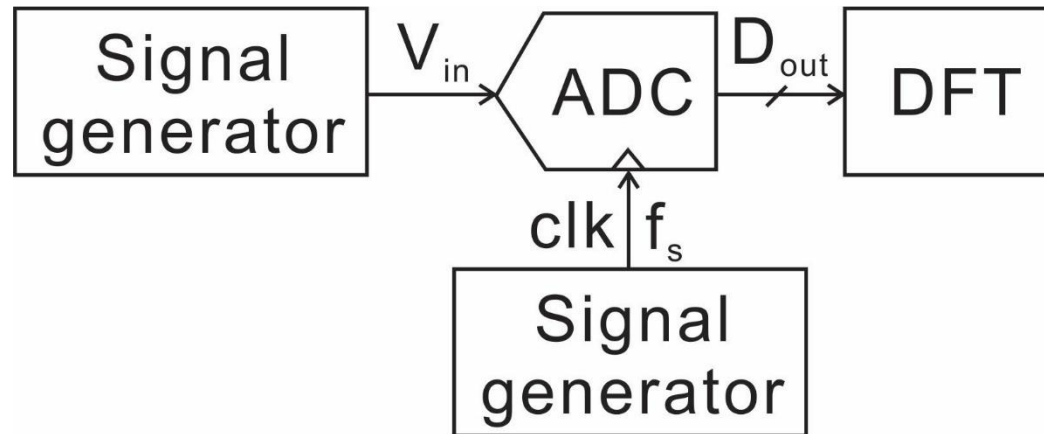
$$\text{INL}[k] = \frac{V_{\text{out}}[k] - V_{\text{out},0}[k]}{\text{Step}_{\text{avg}}}$$

- $V_{\text{out},0}[k]$ is the ideal output voltage for code k

Spectral Metrics

- Static metrics cannot capture dynamic errors.
- ADC/DAC performance is a function of frequency.
- Effective performance evaluation in the frequency domain
 - Apply sine waves at the input and look at the output spectrum
 - Expect same sine tones at the output, all other frequency components represent nonidealities
- List of spectral metrics:
 - SNR: Signal-to-noise ratio
 - SNDR (SINAD): Signal-to-(noise + distortion) ratio
 - ENOB: Effective number of bits
 - DR: Dynamic range
 - SFDR: Spurious free dynamic range
 - THD: Total harmonic distortion
 - ERBW: Effective Resolution Bandwidth
 - IMD: Intermodulation distortion
 - MTPR: Multi-tone power ratio

ADC Spectral Metrics Test Setup



- What are the differences between DFT, CTFT, and DTFT?

Fourier Transforms (1)

- Continuous-time FT:

$$X(\omega) = \int_{-\infty}^{\infty} x(t)e^{-j2\pi\omega t} dt$$

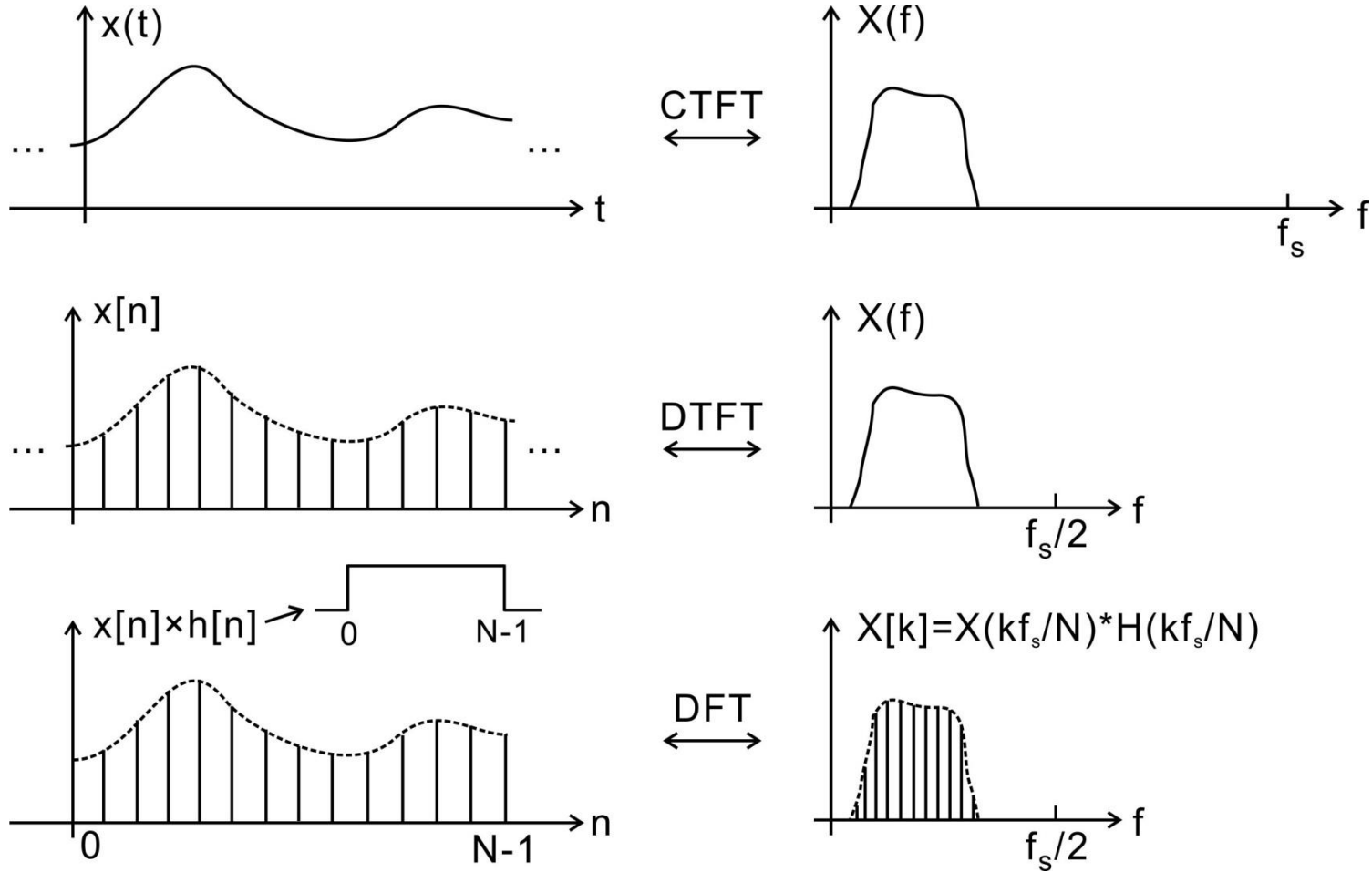
- Discrete-time FT (Z transform):

$$X(\omega) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

- Discrete FT:

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N}$$

Fourier Transforms (2)



- $h[n]$ is a rectangular window. Is there any distortion in the spectrum?

Discrete Fourier Transform

- DFT takes a block of N time-domain samples (spaced $T_s=1/f_s$) and produces a set of N frequency bins

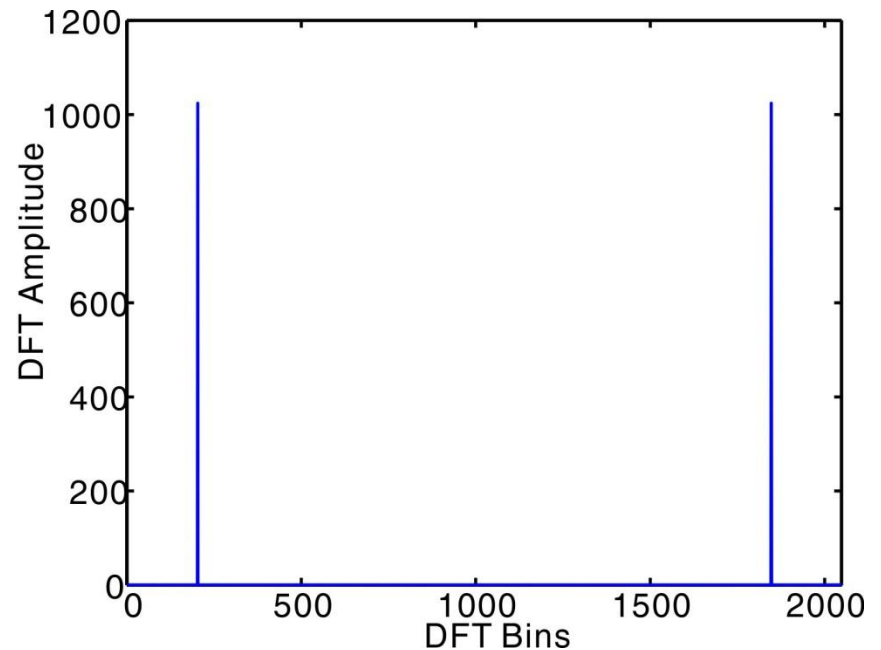
$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}$$

- Bin k represents frequency content at $k \cdot f_s/N$
- DFT frequency resolution
 - Proportional to $f_s/N = 1/(N \cdot T_s)$
 - $N \cdot T_s$ is total time spent on gathering samples
- Parseval's theorem:

$$\sum_{n=0}^{N-1} |x[n]|^2 = \frac{1}{N} \sum_{k=0}^{N-1} |X[k]|^2$$

Direct DFT Plot

- $N = 2048$; $f_s = 1024$; $f_{in} = 101$;
- $D_{out} = V_{in}$;
- $D_{out_fft} = \text{abs}(\text{fft}(D_{out}))$;
- Drawbacks
 - Amplitude depends on N
 - Do not know the frequency
 - Cannot clearly see the spectrum other than the fundamental tones
 - Have two identical half

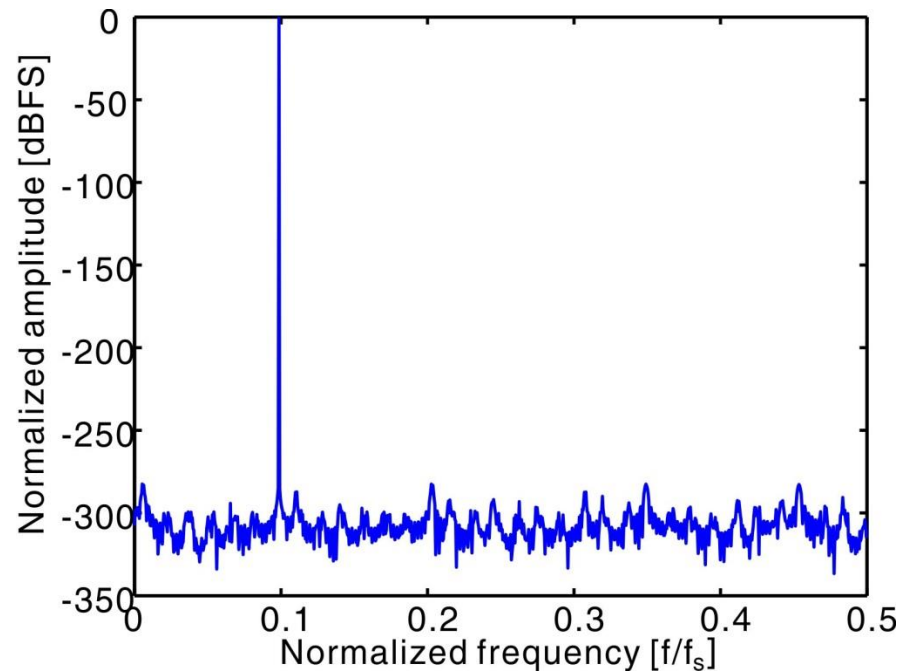


Normalized Plot

Improvements:

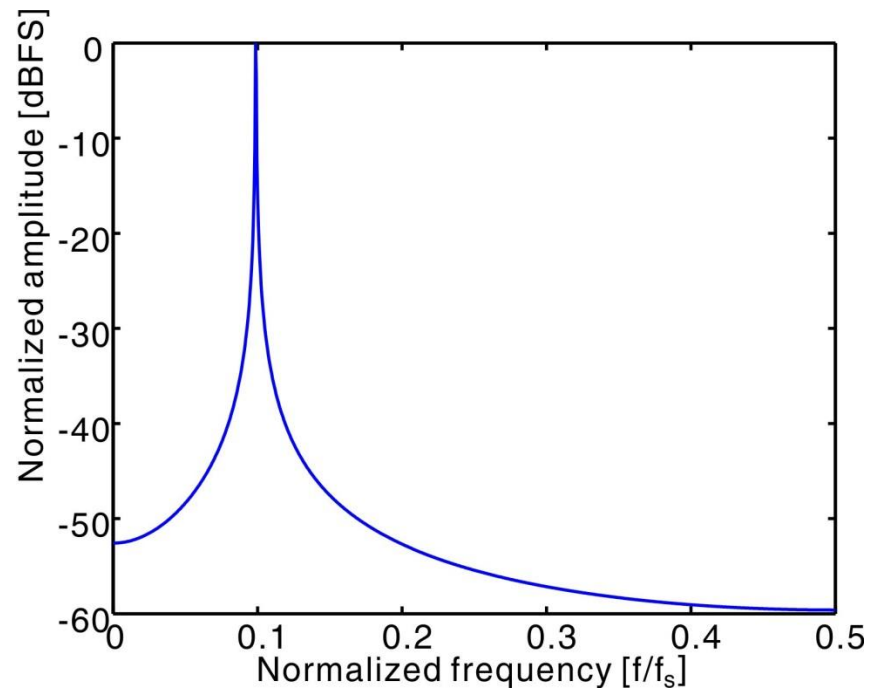
- Normalize the amplitude [dBFS]
- Normalize the frequency [f/f_s]
- Plot the amplitude in log scale
- Remove the redundant half

Question: why there is a noise floor?

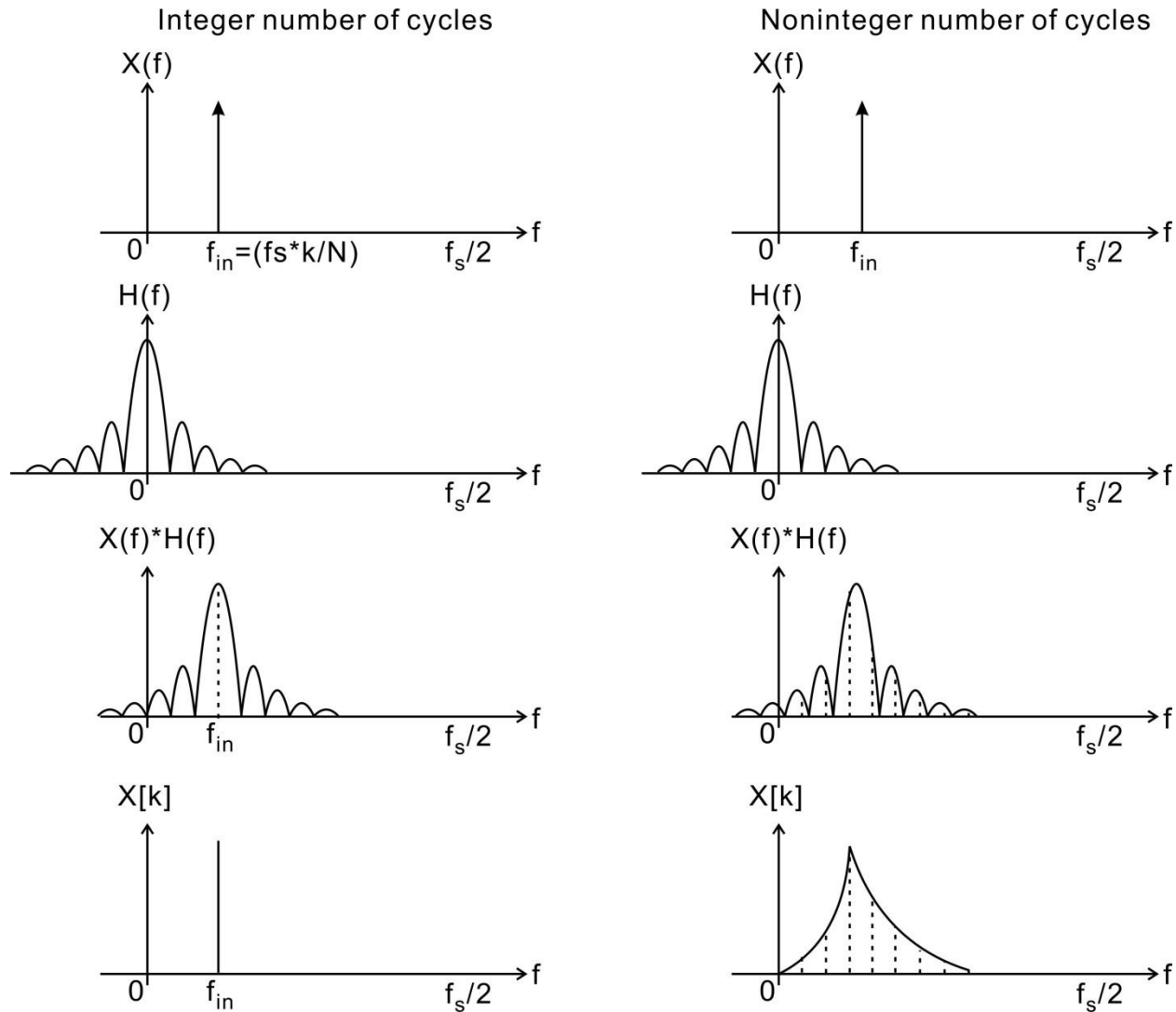


Change Frequency

- All things remain the same except $f_{in} = 101 \rightarrow 101.2$
- What's wrong?



Spectral Leakage

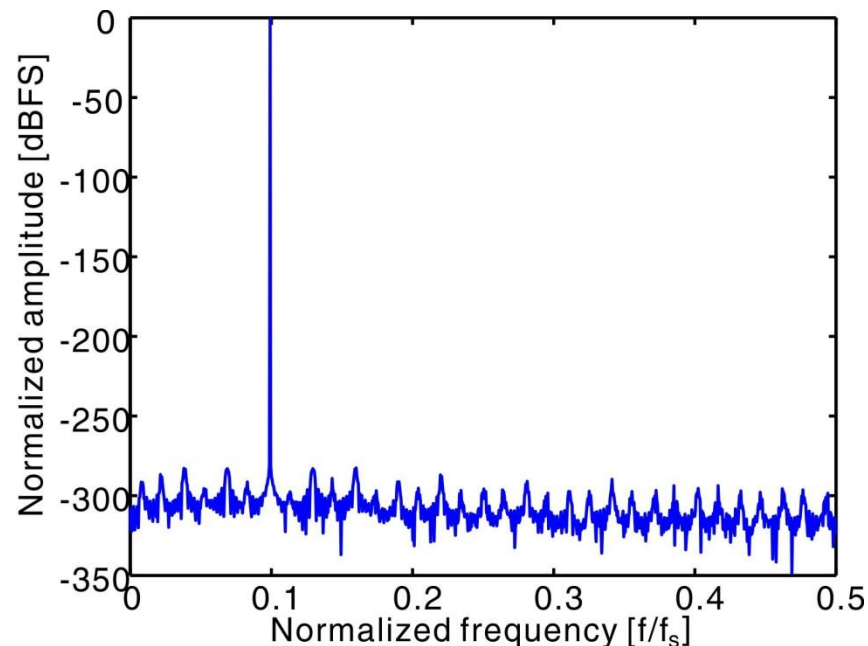


Summary

- DFT is the sampled spectrum of $x[n] \times h[n]$
 - $h[n]$ is a rectangular window.
- If $x[n]$ is not periodic of N , there is spectral leakage.
- 2 ways to solve this problem
 - Ensure integer number of cycles
 - Use other window functions with smaller side lobes

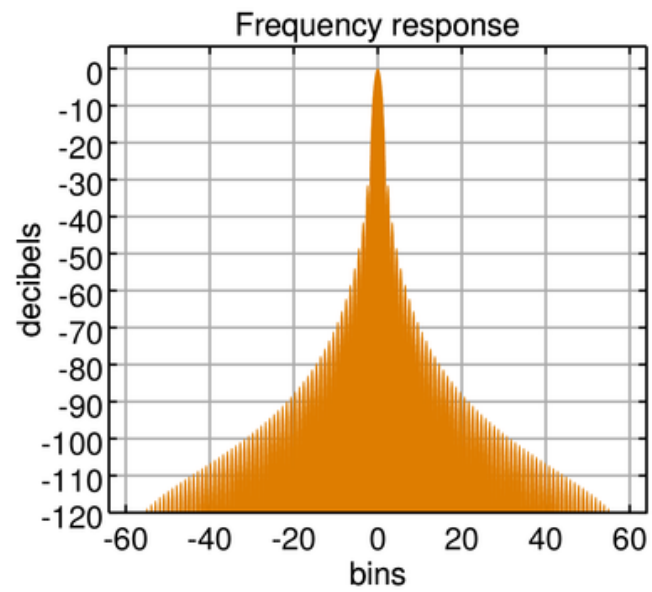
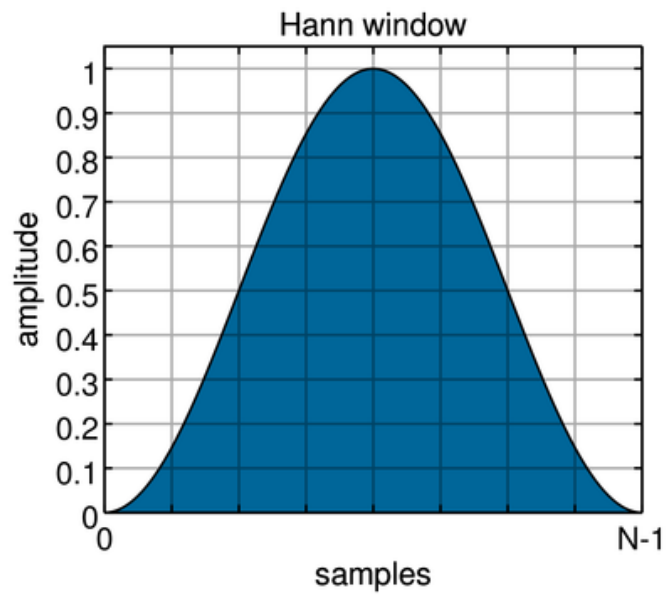
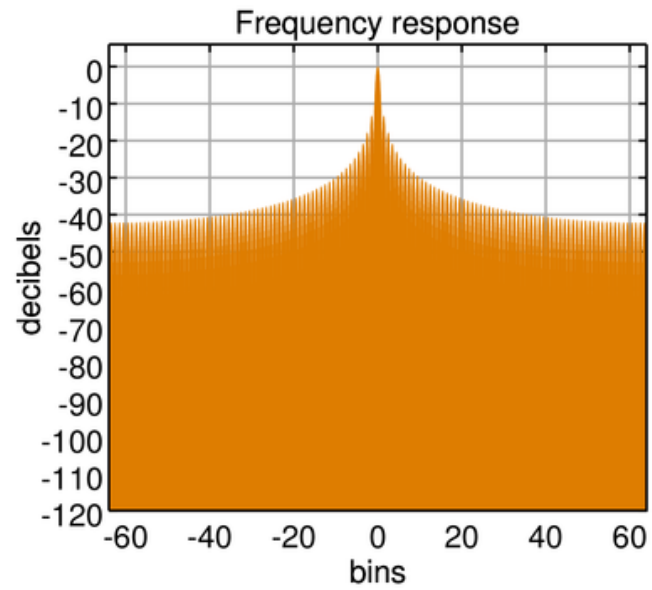
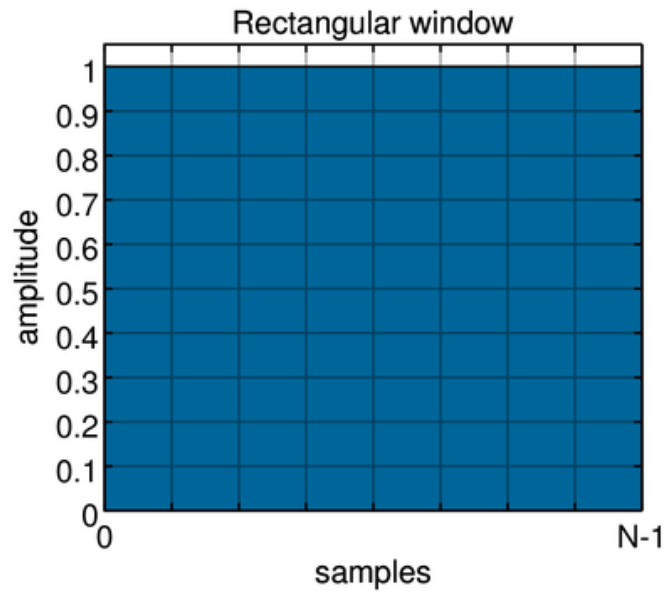
Ensure Integer Number of Cycles

- $N = 2048$; $f_s = 1024$;
- $f_{in} = f_s/N \cdot \text{cycle}$;
- cycle must be an integer
 - E.g., cycle = 203;
 - $f_{in} = f_s/N \cdot \text{cycle} = 101.5$
- How many usable frequencies are there in $(0, f_s/2)$?
 - $0 < \text{cycle} < N/2$

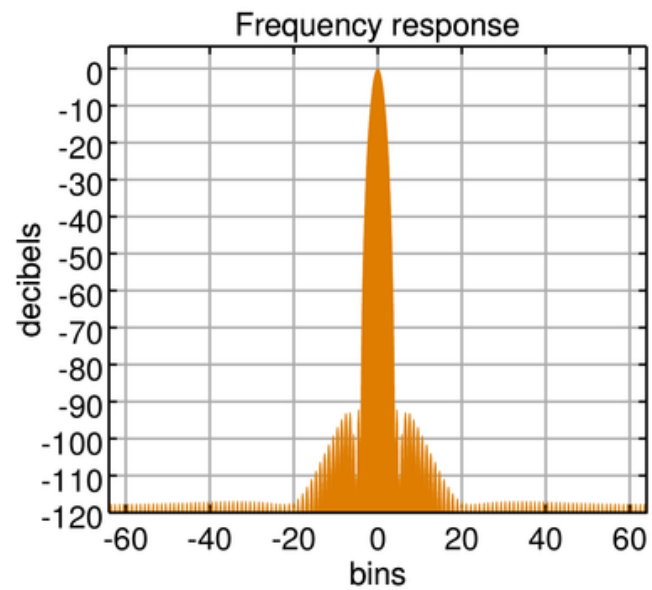
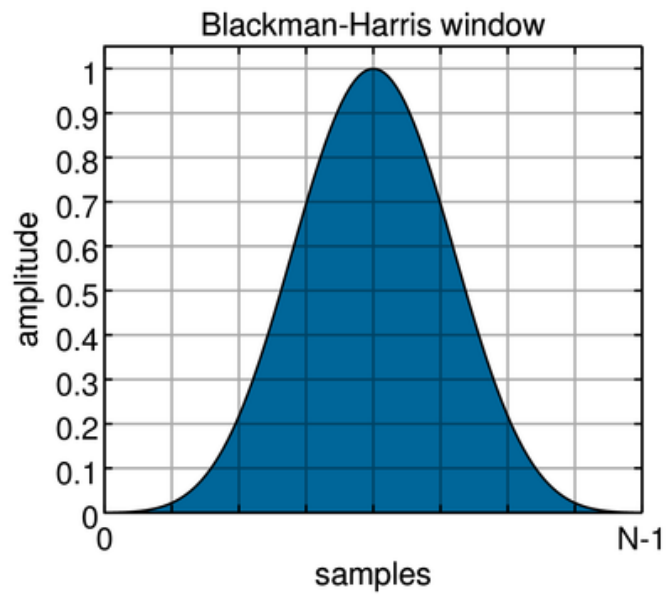
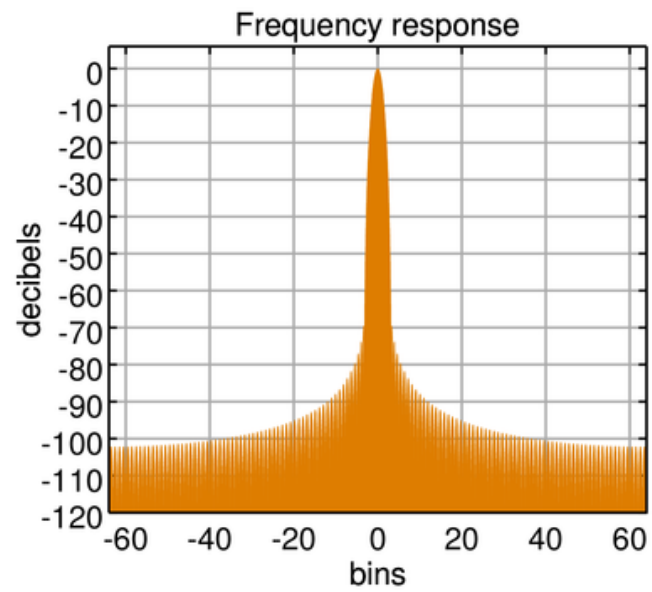
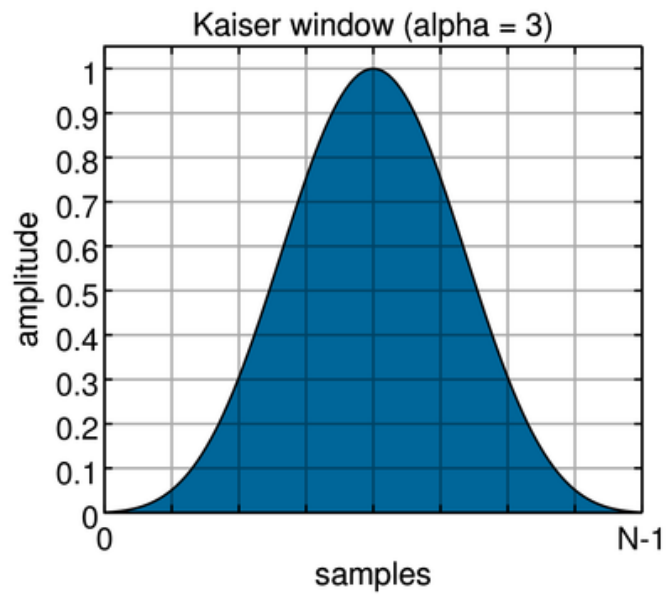


Windowing

- Use a different window function with smaller side lobes than that of the rectangular window
- Many window functions to choose from:
 - Hann, Hamming, Blackman, Kaiser, Cosine, Hanning, ...
 - Tradeoff: main-lobe width vs. side-lobe attenuation
- The width of the main-lobe is now larger
 - The signal spans more-than-1 frequency bins

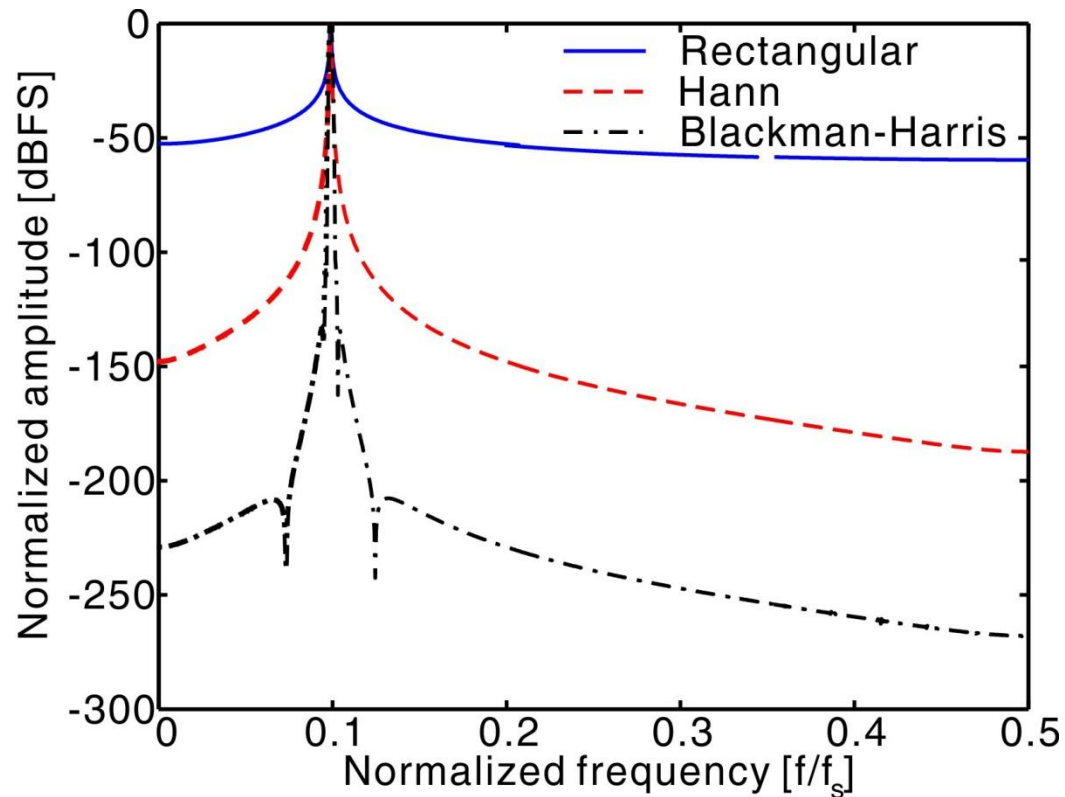


http://en.wikipedia.org/wiki/Window_function



http://en.wikipedia.org/wiki/Window_function

Spectrum with Windowing



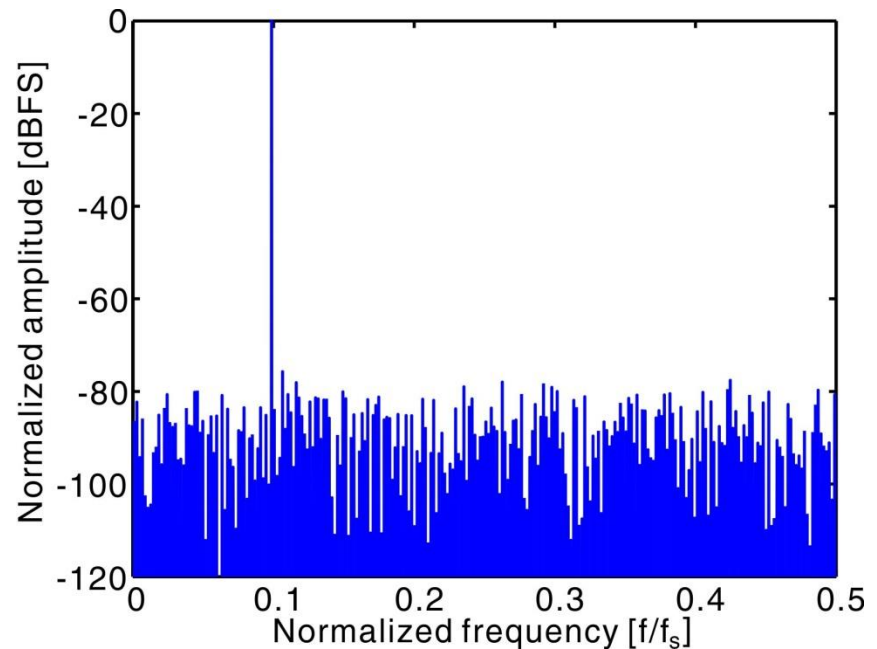
Dout = Dout.*window_func

Integer Cycles vs. Windowing

- Integer number of cycles
 - Input falls into a single DFT bin 😊
 - No smearing
 - Highest spectral resolution
 - Requires careful choice of the input frequency 😞
 - Ideal for simulations
 - In real measurements, we can set the input signal to be a fraction of the clock signal
 - “Coherent sampling”
- Windowing
 - No restrictions on signal frequency 😊
 - Signal and harmonics distributed over several DFT bins 😞
 - Be careful about the smeared out nonidealities
 - Requires more samples for a given accuracy

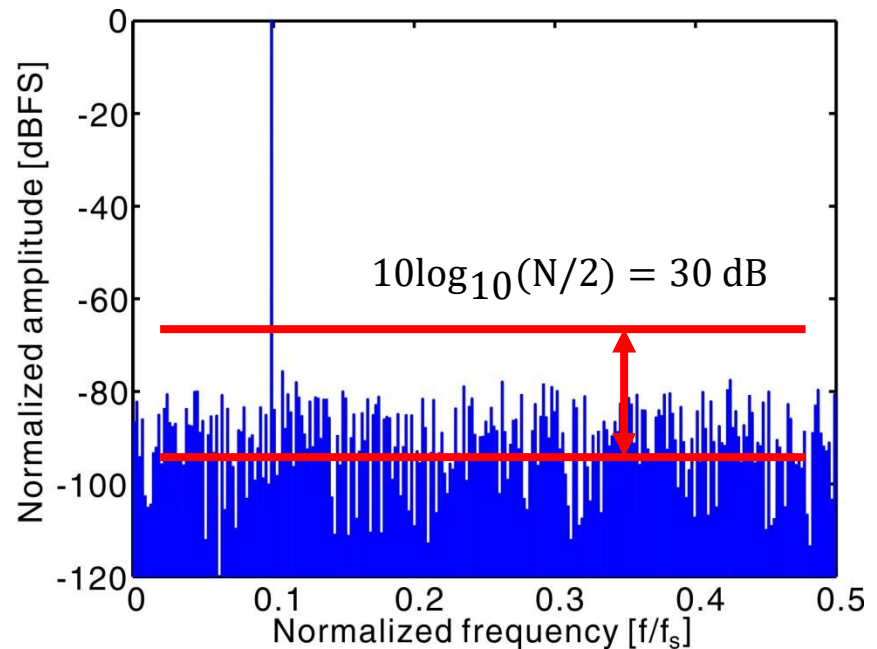
Spectrum with Quantization Noise

- $N = 2048$; $f_s = 1024$; $f_{in} = 101$;
- $B = 10$;
- $LSB = 2/2^B$;
- $D_{out} = \text{round}(V_{in}/LSB) * LSB$;
- Noise spectrum looks flat
- How to calculate SQNR from FFT?
 - Use Parseval's theorem
 - Time-domain power = frequency-domain power
 - $SQNR = 6B + 2 = 62 \text{ dB}$
- Why noise floor is not at -62 dBFS?



Noise Floor

- Total quantization noise is distributed in all fft bins
- The total noise is -62 dBFS, and thus, the noise floor (which is the noise in each bin) is $-62 \text{ dBFS} / (N/2) = -92 \text{ dBFS}$
- The noise floor depends on N , and thus, the total noise cannot be inferred if N is not specified.

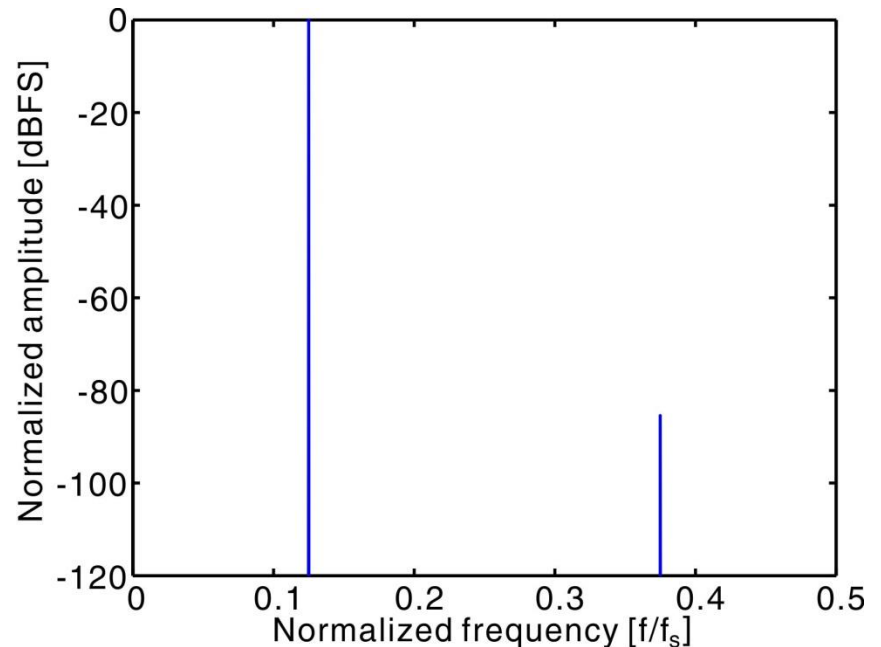


DFT Plot Annotation

- Most common annotation
 - Specify N: how many DFT points were used
- Less common options
 - Shift DFT noise floor by $10\log_{10}(N/2)$ dB
 - Normalize the bin width to 1 Hz and express noise as power spectral density (similar to continuous-time FT)
 - "Noise power in 1 Hz bandwidth"

Periodic Quantization Noise

- $f_{in} = 101 \rightarrow 128$
- $f_{in} = f_s / 8$
- Quantization noise is repetitive and highly correlated with the input.
- To make noise spectrum flat and span all bins, it is important to make N and cycle mutually prime
 - $\text{GCD}(N, \text{cycle}) = 1$



Typical ADC Output Spectrum

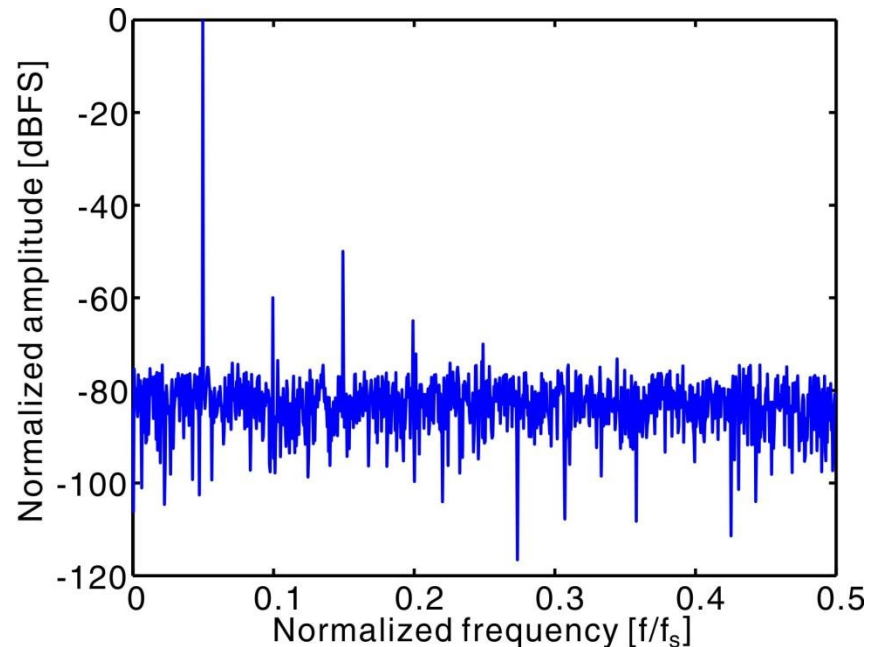
- Harmonics due to nonlinearities
- Signal-to-distortion ratio (SDR)

$$\text{SDR} = \frac{\text{Signal Power}}{\text{Total Distortion Power}}$$

- Total harmonic distortion

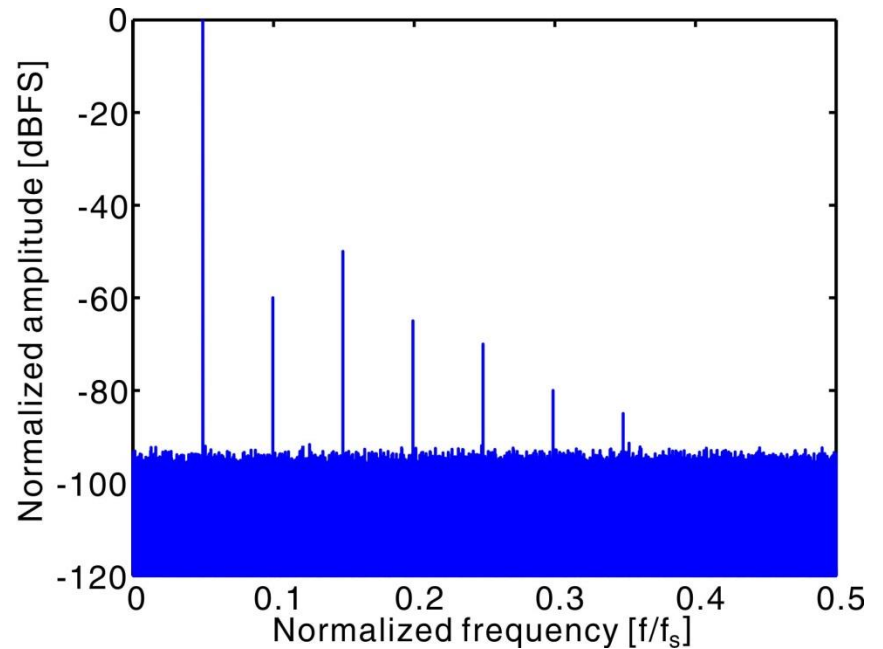
$$\text{THD} = \frac{1}{\text{SDR}}$$

- By convention, total distortion power consists of 2nd through 7th harmonics
 - Other distortion induced harmonics are ignored, why?
- Are there 6th and 7th harmonics?



Lowering the Noise Floor

- We can lower the noise floor by increasing the FFT size.
- $N = 2^{11} \rightarrow 2^{18}$
- We clearly see all harmonics.

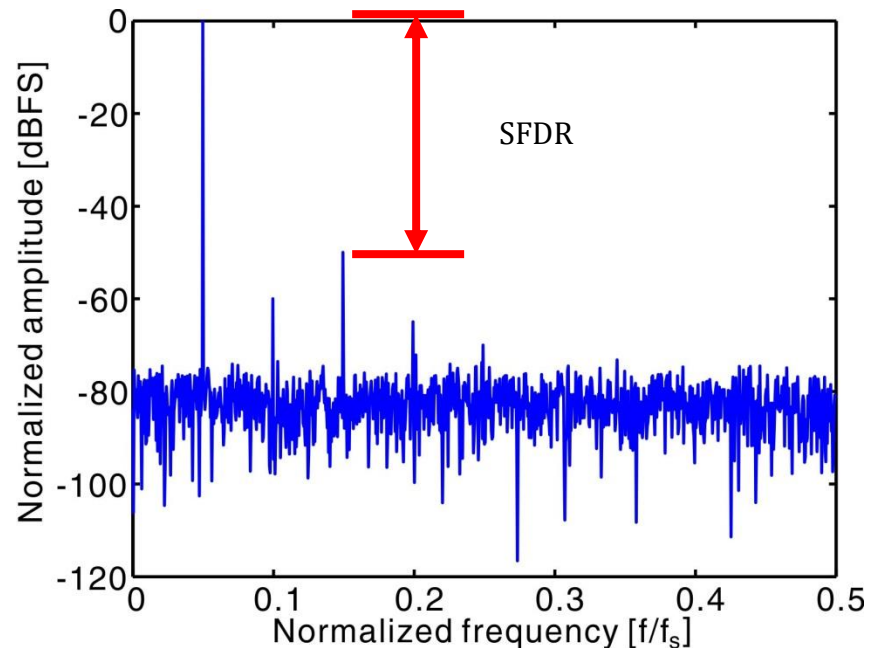


SFDR

- Spurious free dynamic range (SFDR)

$$\text{SFDR} = \frac{\text{Signal Power}}{\text{Largest Spurious Power}}$$

- Largest spur is often, but not necessarily, an input harmonic
- Need to specify the input signal power, why?

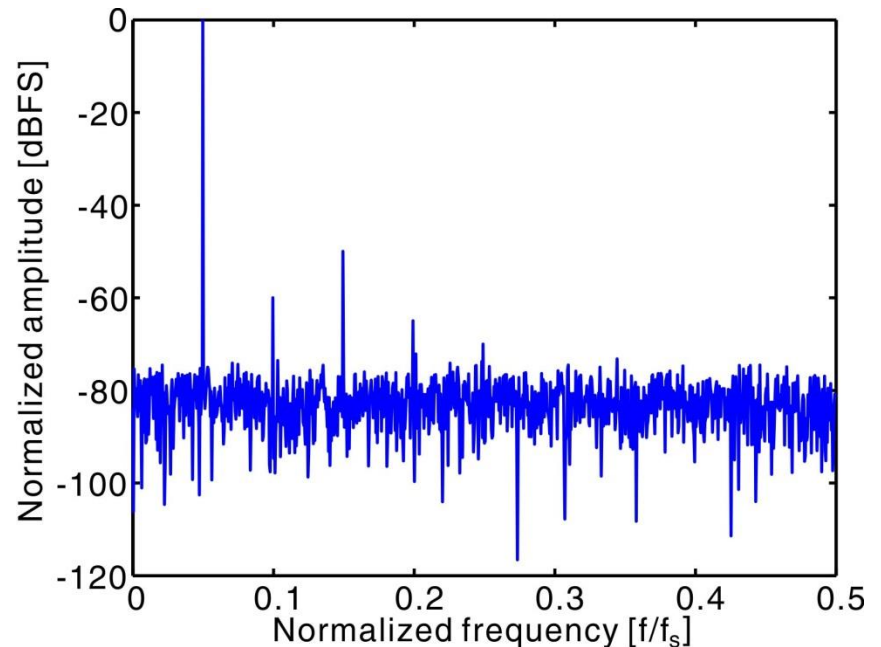


SNR

- Signal-to-noise ratio (SNR):

$$\text{SNR} = \frac{\text{Signal Power}}{\text{Total Noise Power}}$$

- Total noise power includes all bins except DC, input, and 2nd through 7th harmonic.
- It consists of:
 - Quantization noise
 - Electronic noise
 - “DNL noise”
 - Other distortion terms
 - Sampling errors



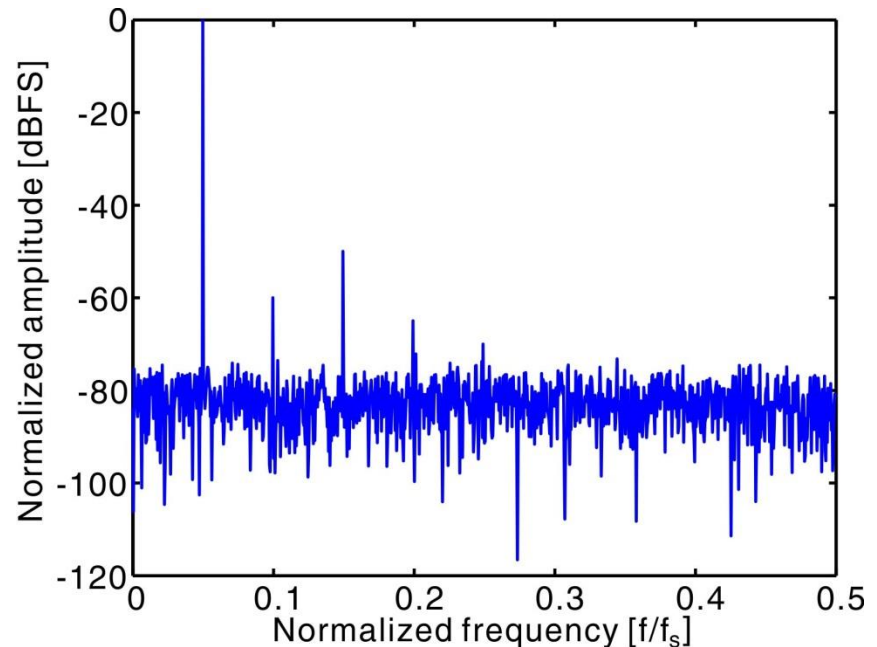
SNDR and ENOB

- Signal-to-(noise + distortion) ratio (SNDR):

$$\text{SNDR} = \frac{\text{Signal Power}}{\text{Noise and Distortion Power}}$$

- Total noise and distortion includes all bins except DC and the input.
- Effective number of bit (ENOB):

$$\text{ENOB} = \frac{\text{SNDR (dB)} - 1.76}{6.02} < B$$



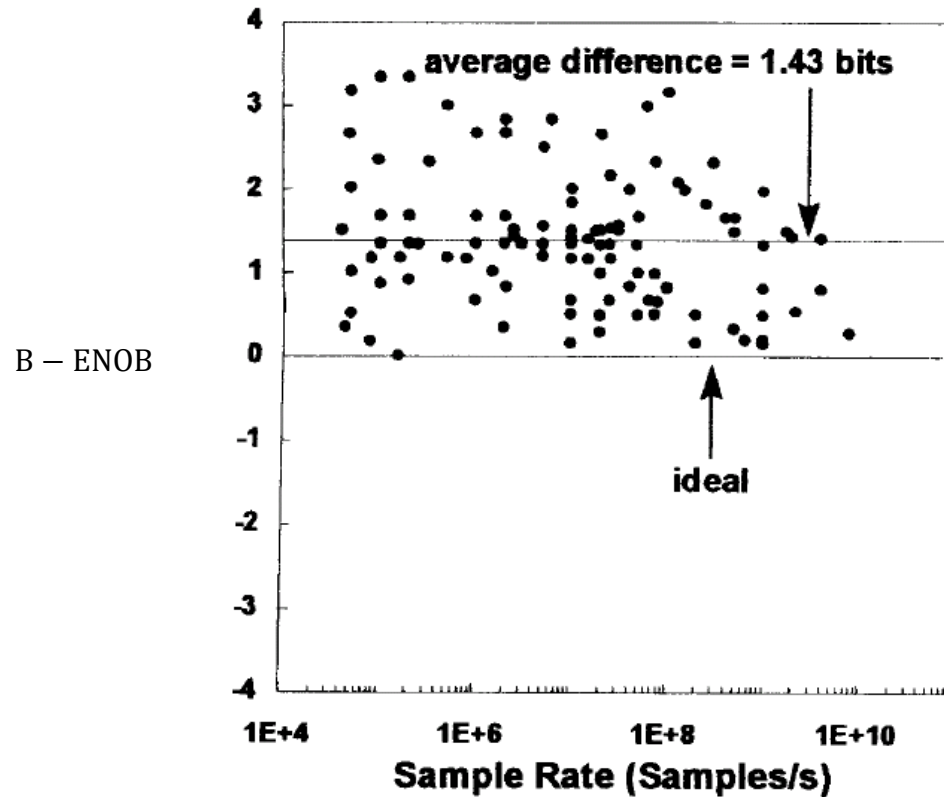
Effective Number of Bits

- What is the ENOB for a 10-bit converter with 50 dB SNDR?

$$\text{ENOB} \cong \frac{50 - 2}{6} = 8$$

- Assuming the stated resolution⁶ of an ADC is B, ENOB = B only for “0” electronic noise and “0” distortion
- Distortion can be lowered through better design.
 - For a well designed ADC, SNDR should not be limited by distortion.
- Electronic noise is more fundamental. It directly trades with power.
- Usually electronic noise is made comparable with the quantization noise.
 - Diminishing return for further reducing electronic noise
- For good power efficiency: ENOB < B-1

Survey Data



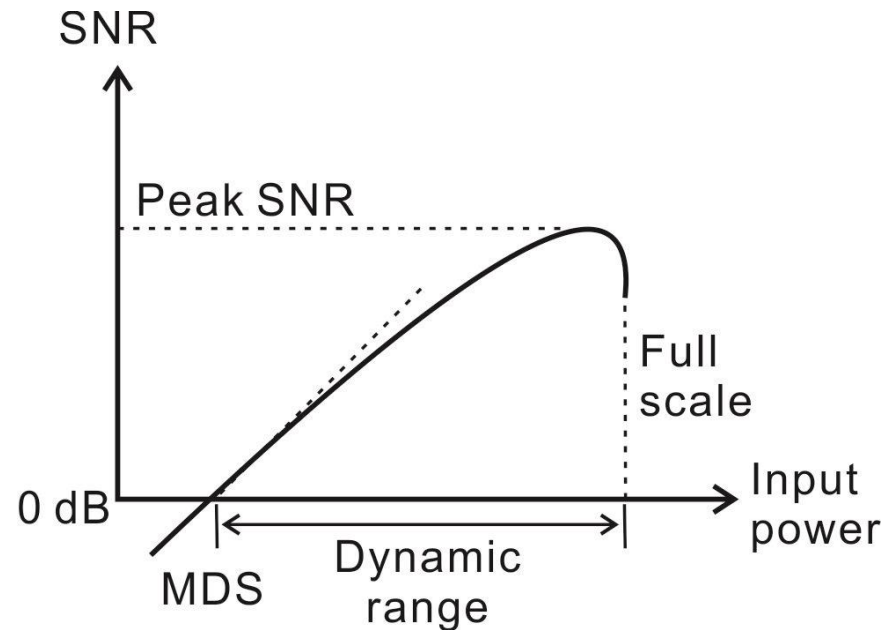
R. H. Walden, "Analog-to-digital converter survey and analysis," *IEEE J. on Selected Areas in Communications*, pp. 539-50, April 1999

Dynamic Range

- Dynamic range:

$$DR = \frac{\text{Maximum Signal Power}}{\text{Minimum Detectable Signal Power}}$$

- Minimum detectable signal (MDS) is defined as the power of the input that produces 0 dB SNR.
- $DR \geq \text{peak SNR}$
 - As signal amplitude increases, the noise floor may rise.



Aliasing

- Due to aliasing, harmonics can appear at frequencies that do not look like harmonics.

- For example:

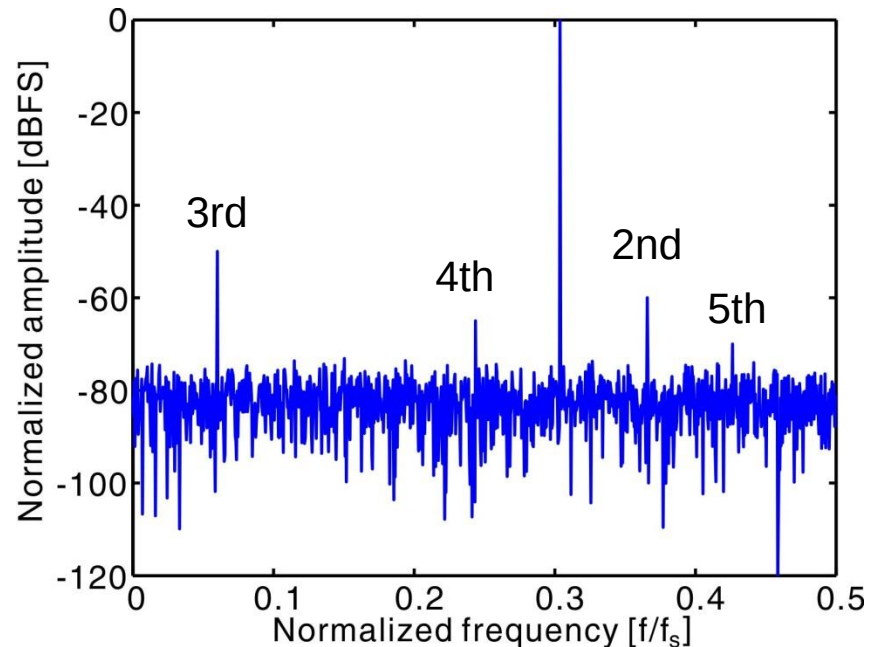
$$f_1 = f_{in} = 0.3125 f_s$$

$$f_2 = 2f_1 = 0.6250 f_s \rightarrow 0.3750 f_s$$

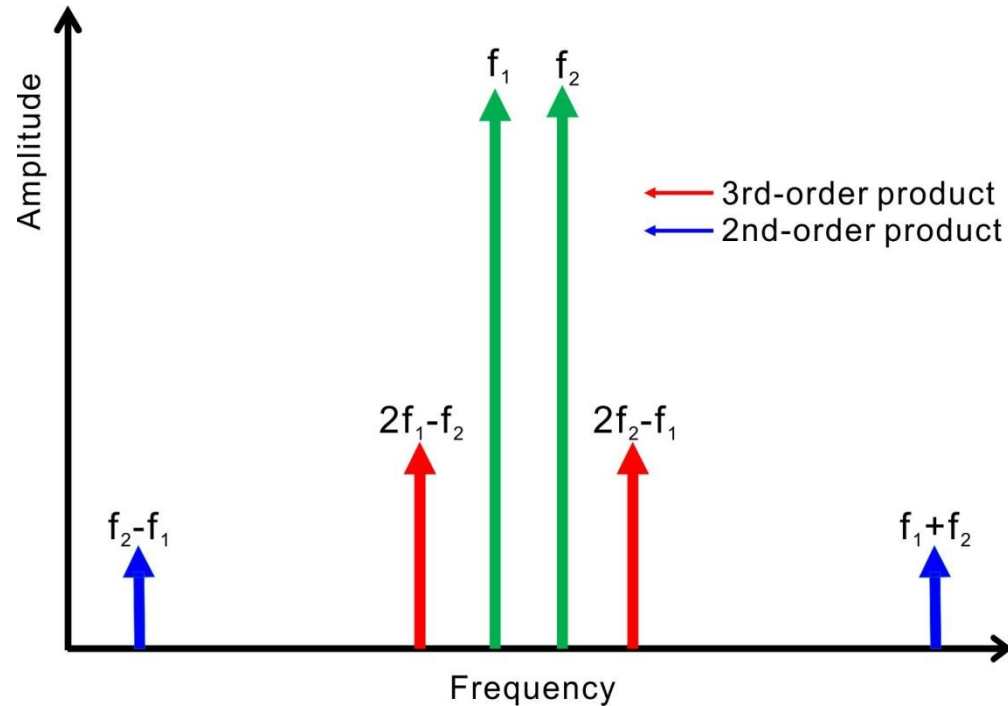
$$f_3 = 3f_1 = 0.9375 f_s \rightarrow 0.0625 f_s$$

$$f_4 = 4f_1 = 1.2500 f_s \rightarrow 0.2500 f_s$$

$$f_5 = 5f_1 = 1.5625 f_s \rightarrow 0.4375 f_s$$

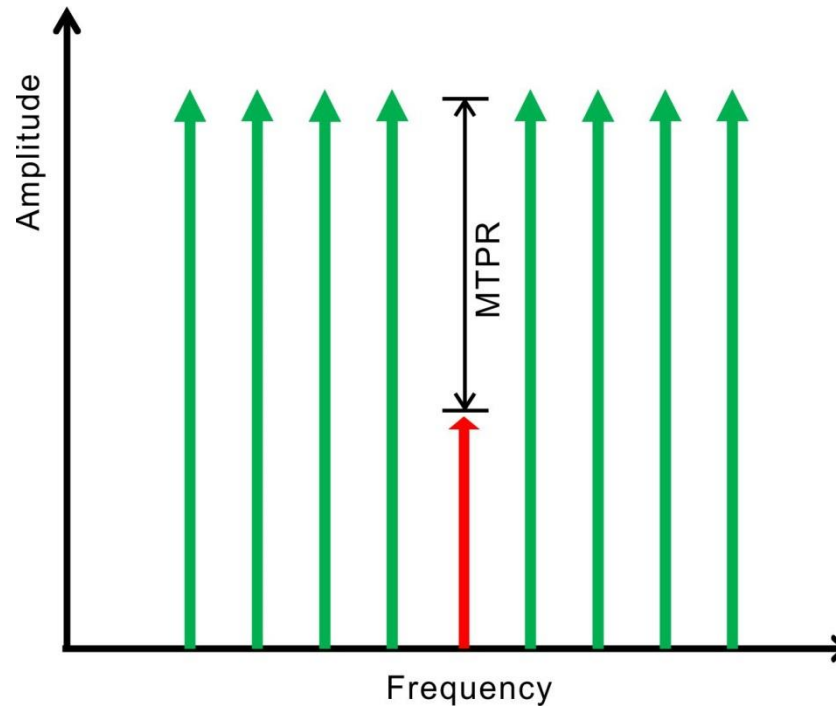


Intermodulation Distortion



- IMD is important in multi-channel communication
 - 3rd-order products are difficult to filter out

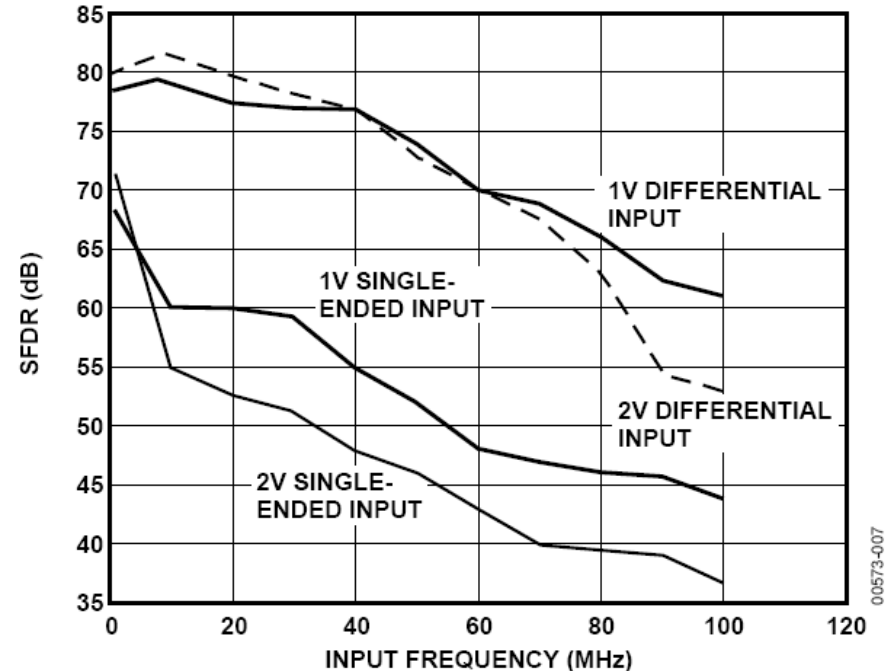
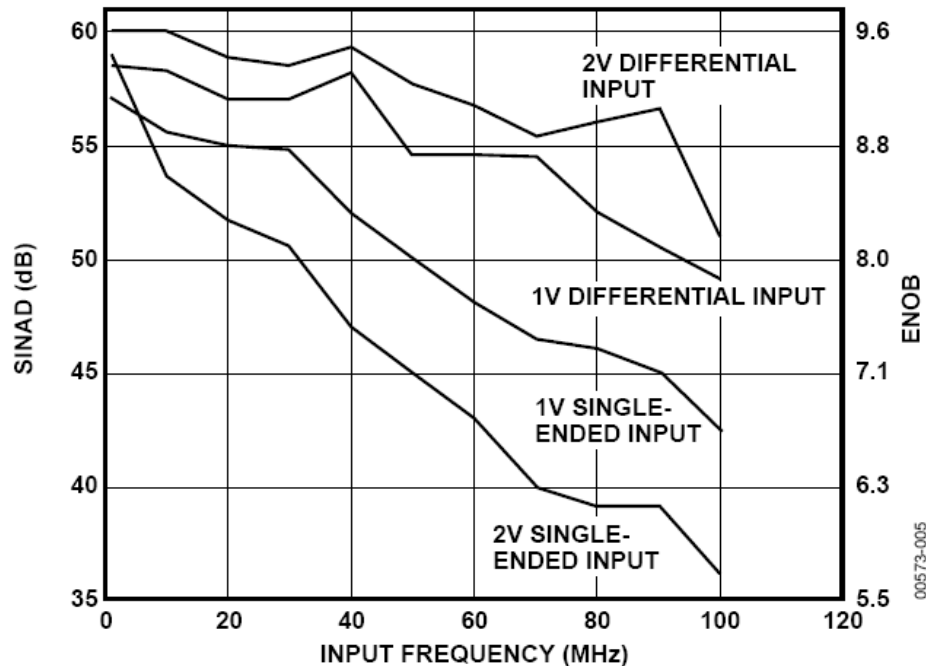
Multi Tone Power Ratio (MTPR)



- MTPR is a useful metric in multi-tone transmission systems
 - E.g. OFDM

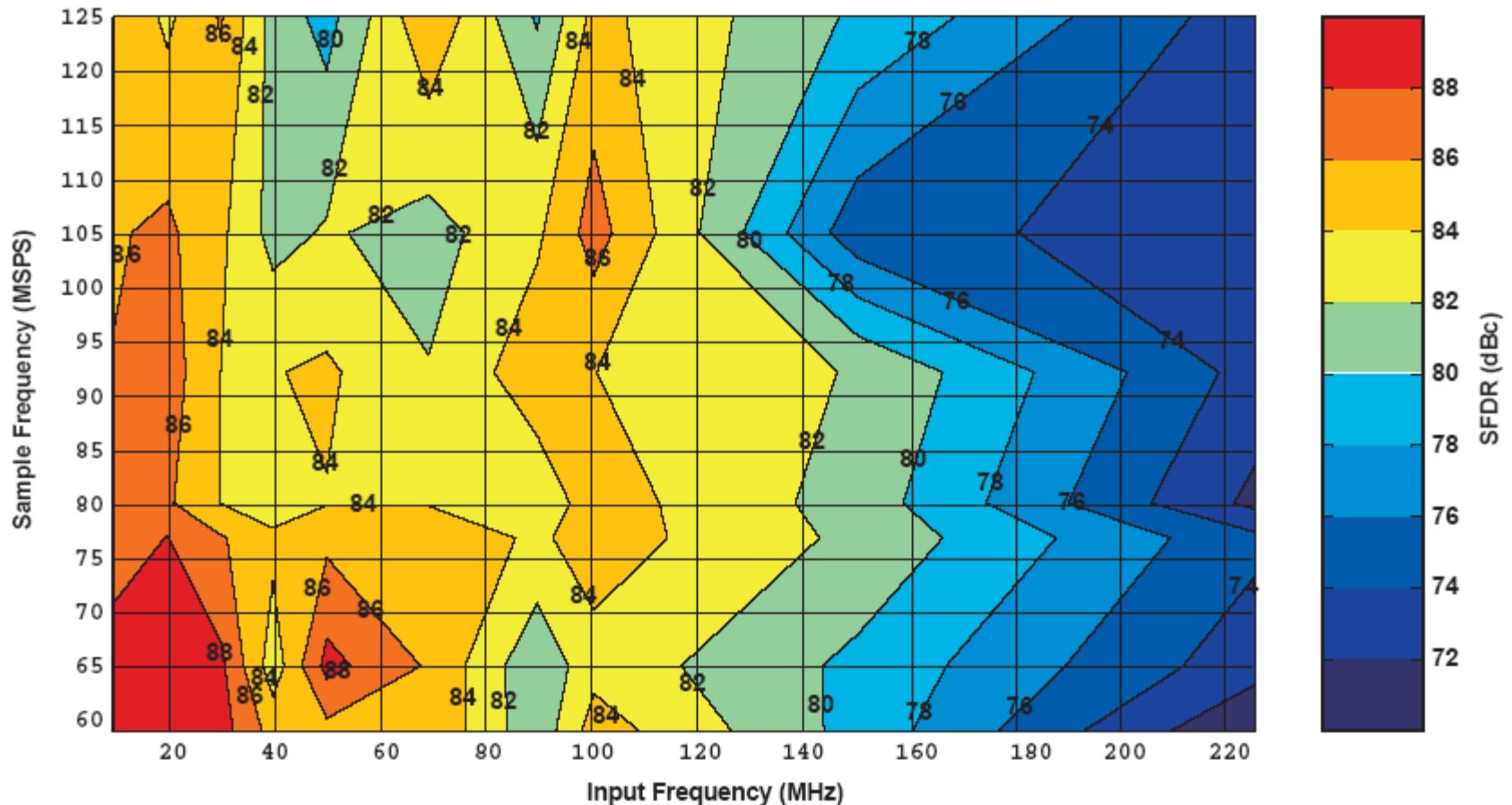
Frequency Dependence (1)

- Data converter performance depends on both the sampling and the input frequency



[Analog Devices, AD9203 Datasheet]

Frequency Dependence (2)

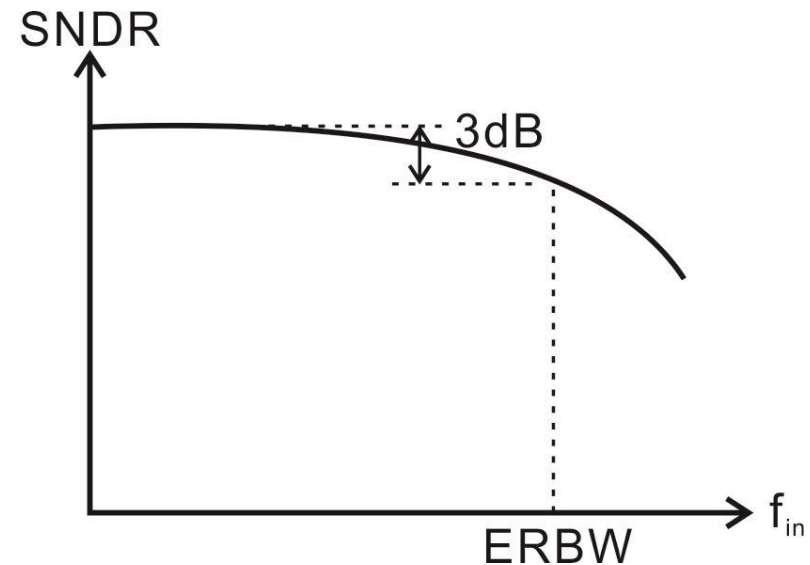


[Texas Instruments, ADS5541 Datasheet]

- An island surrounded by ocean

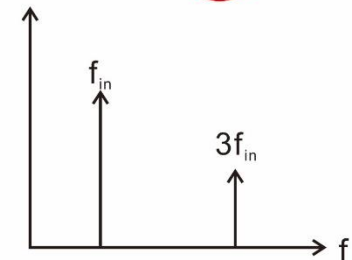
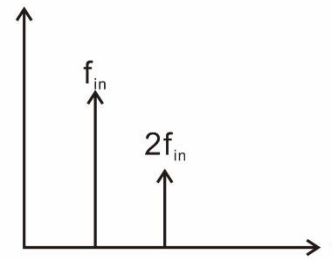
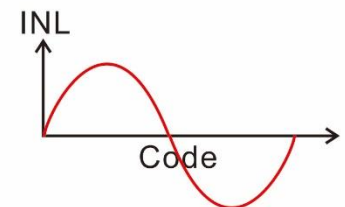
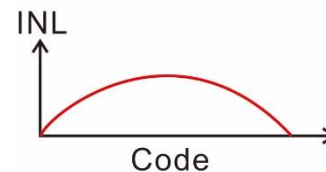
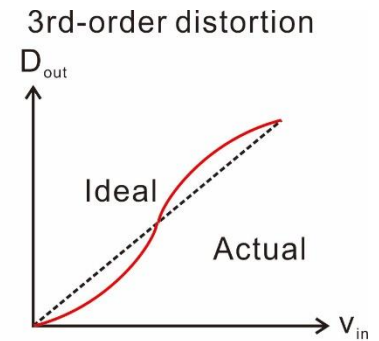
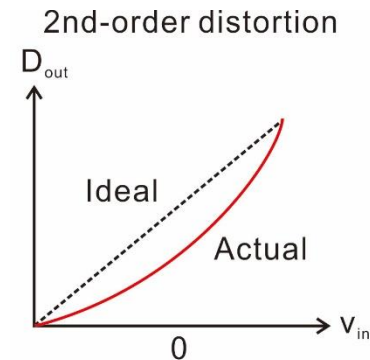
Effective Resolution Bandwidth (ERBW)

- ERBW is the input frequency at which the SNDR drops by 3dB compared to low frequency
 - 0.5-bit loss in ENOB
- $\text{ERBW} \approx f_s/2$ for a well designed Nyquist rate ADC
- $\text{ERBW} > f_s/2$ is not uncommon, especially for subsampling ADCs



INL/DNL and SFDR/THD/SNR

- Static nonlinearities (INL/DNL) and spectral nonlinearities (SFDR /THD/SNR) are closely related especially at low frequencies
 - At high frequencies, spectral nonlinearities tend to be limited by dynamic errors
- Can link systematic INL patterns with harmonics



SNR Degradation due to DNL

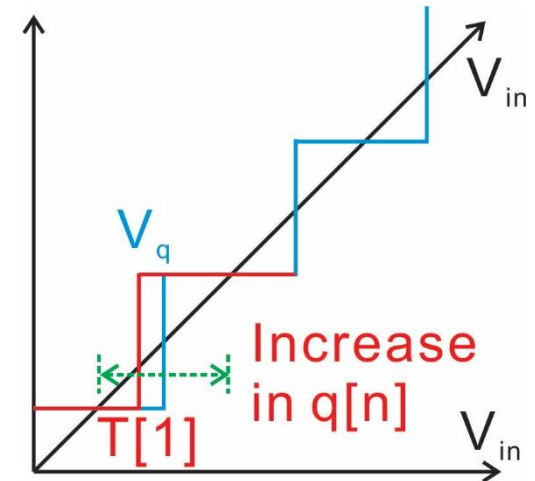
- If INL and DNL show irregular patterns, the nonlinearities may show up as “noise” like, which causes SNR degradation.
- Changes in the transition levels cause quantization noise to increase.
- For simplicity, let us consider a 1-bit quantizer with range of $[-1,+1]$. The transition level is changed from 0 to δ . As a result, the quantization noise is increased to:

$$\sigma^2 = \frac{1}{2} \left(\int_{\delta}^1 (V_{\text{in}} - 0.5)^2 dV_{\text{in}} + \int_{-1}^{\delta} (V_{\text{in}} + 0.5)^2 dV_{\text{in}} \right) = \frac{1}{12} + \frac{1}{2} \delta^2$$

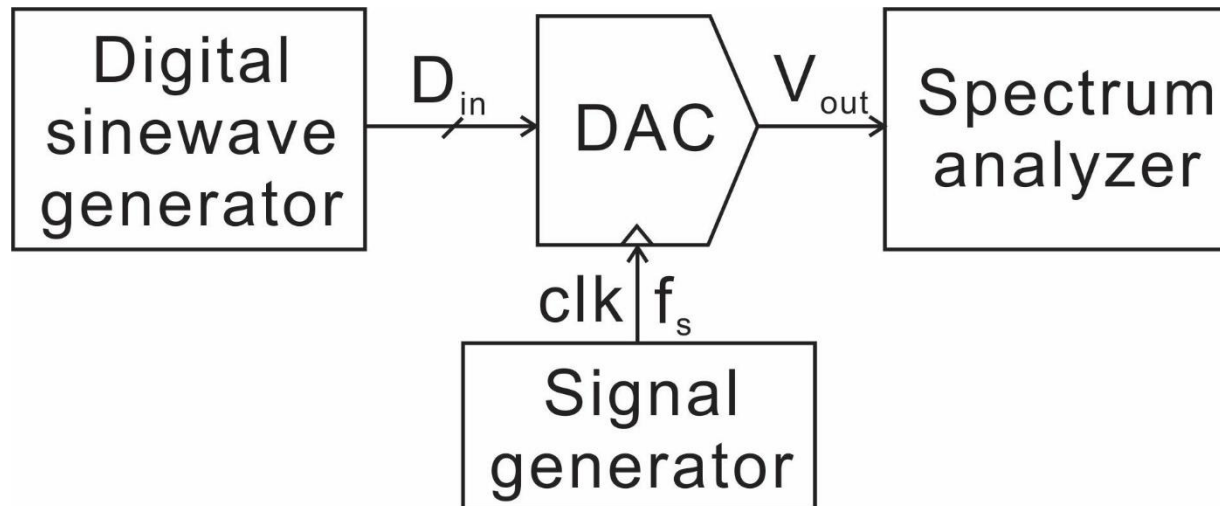
- Thus, if $\delta = 0.5$, the quantization noise is more than doubled.
- For multibit quantization, assuming that transition-level deviation δ is independent and uniformly distributed in $[-0.5,+0.5]$ LSB, $E\delta^2 = 1/12$. Thus, the amount of quantization noise increase about 50% or 1.8 dB
- This also translates to a DNL with triangular pdf distribution in $[-1,+1]$

Transition-level, INL, and DNL noise

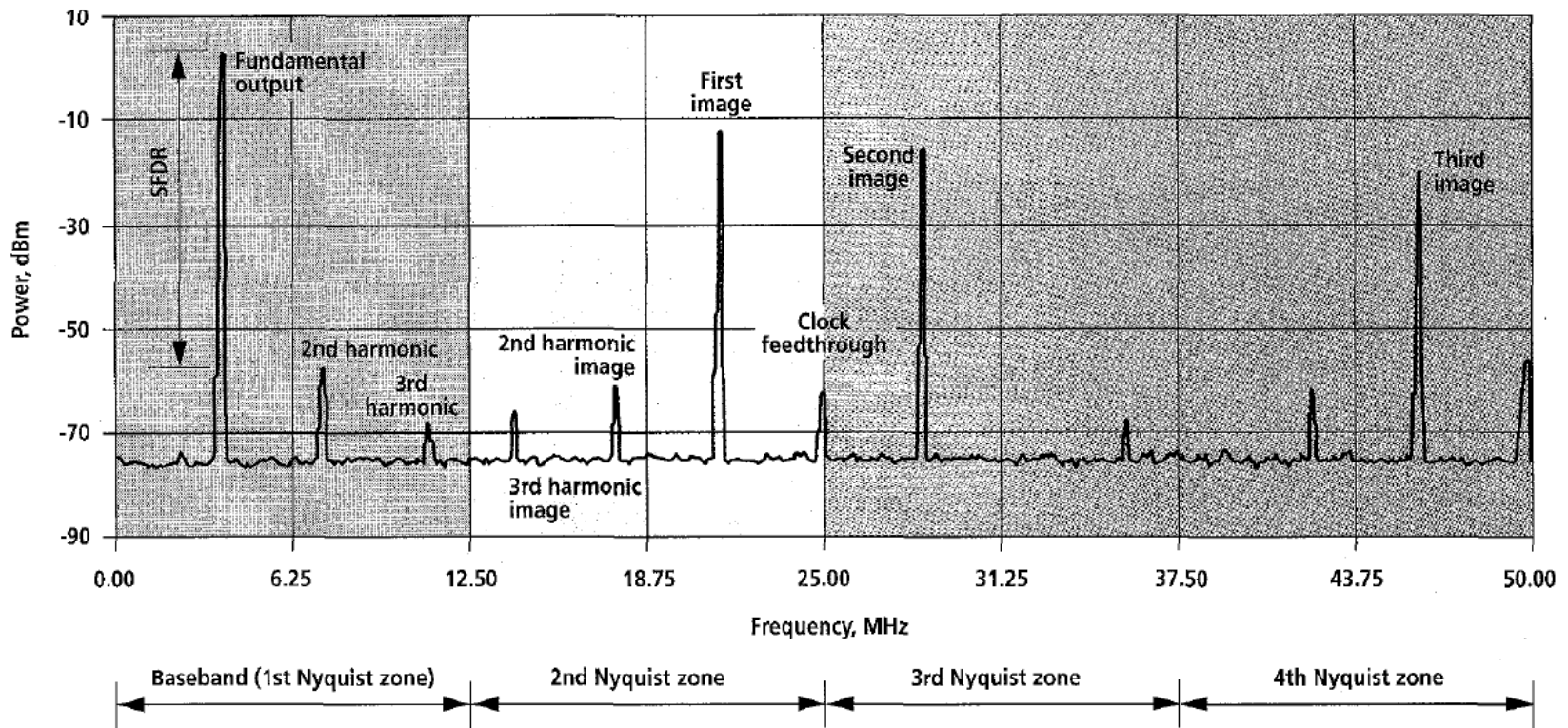
- Transition levels and INL are directly linked
- If we assume certain distribution of transition levels, it directly determines the distribution of INL.
 - For example, if we assume that transition level deviation δ follow independent and uniform distribution of $[-0.5, +0.5]$ LSB, it directly means that INL follows the same distribution, because INL is by definition the transition level deviations.
 - $DNL[i] = T[i+1] - T[i] - 1 = \delta[i+1] - \delta[i]$
 - Thus, DNL follows triangular distribution from -1 to 1 LSB.



DAC Test Setup



Typical DAC Output Spectrum



[Hendriks, "Specifying Communications DACs, IEEE Spectrum, July 1997]